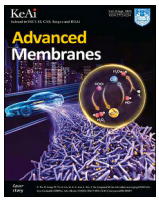
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Hybrid models, digital twins, and digital shadows for sustainable membrane technologies: A critical review

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ABSTRACT

Membrane separation technologies are essential for water purification, wastewater treatment, gas separation, hydrogen (H_2) production, and carbon (CO & CO_2) capture. Despite their wide applicability, large-scale adoption is limited by significant technical challenges, including fouling, the permeability–selectivity trade-off, high energy consumption, and the durability required for long term operation. Integrating machine learning (ML) and artificial intelligence (AI) with the physics-based models, as a hybrid mode, digital shadows (DSs), and digital twins (DTs), offers a promising pathway to overcome these limitations. Hybrid modeling combines physics-based simulations with data-driven learning to produce fast and accurate predictive tools for membrane processes. DSs extend this capability by creating passively updated virtual representations of membrane systems from real-time data, thereby enabling soft sensing, performance forecasting, and anomaly detection. DTs further advance these functions through bi-directional connectivity and higher-fidelity physical models, supporting adaptive control, predictive maintenance, and system-level optimization. This review examines recent applications of hybrid models, DSs, and DTs across major membrane technologies, including desalination and water treatment, membrane bioreactors, gas separation for CO_2 and H_2, O_2/N_2 separation, catalytic membrane reactors (CMRs), and proton-exchange membrane fuel cells. The role of computational fluid dynamics (CFD) in generating synthetic data, enhancing mechanistic understanding, and supporting model development is also discussed. Case studies demonstrate improvements in fouling prediction, energy-efficient operating set points, reduced operational costs, and real-time decision support at both module and plant scales. Collectively, hybrid modeling enables high-fidelity predictions, DSs support continuous monitoring, and DTs provide a pathway toward intelligent, self-optimizing, and sustainable membrane-based separation systems, bridging the gap between laboratory innovation and industrial deployment.

1. Introduction

Membrane technologies have rapidly advanced and become vital across multiple industries due to their efficiency, simplicity, and adaptability. As selective barriers, membranes enable applications such as water purification, gas separation, H_2 production, and CO_2 capture (Fig. 1) [1,2]. In water treatment, processes including reverse osmosis (RO), nanofiltration (NF), ultrafiltration (UF), and microfiltration (MF), membranes have transformed purification by removing salts, contaminants, suspended solids, and pathogens, thereby improving access to potable water in regions facing scarcity [2–4]. In gas separation, membranes provide an efficient alternative to absorption, adsorption, and cryogenic distillation, relying on selective permeation to simplify applications such as natural gas sweetening, air separation, and CO_2 capture [5,6]. Particularly, membrane systems for CO_2 capture have drawn extensive attention due to their potential in efficiently mitigating greenhouse gas emissions, thereby supporting global efforts in climate change mitigation [7]. Furthermore, membrane technologies significantly enhance H_2

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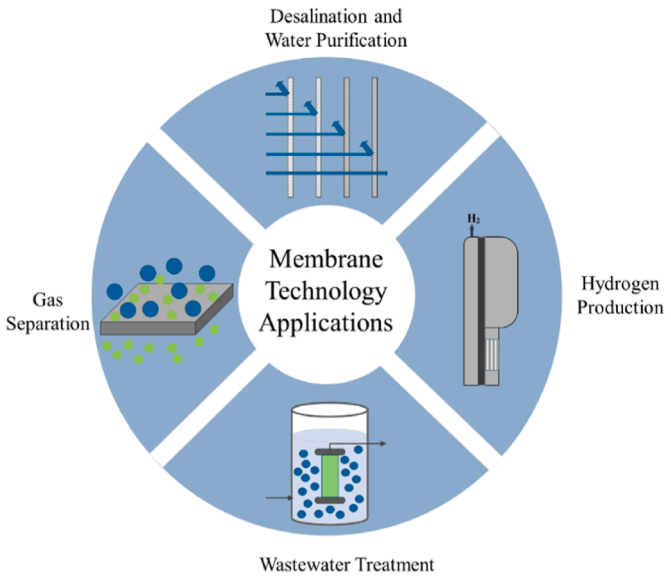


Fig. 1. Applications of membrane processes.

production, especially through membrane reactors (MRs) that integrate reaction and separation to improve H_2 yield, purity, and overall energy efficiency in reforming and gasification processes [3,8].

Therefore, membrane engineering has emerged as a transformative technology across multiple industrial sectors, including chemical and petrochemical processing, biotechnology, energy, and environmental management. Its widespread adoption is driven by inherent advantages, including high selectivity and permeability, low energy requirements, simple operation, scalability, modularity, and ease of process integration. These characteristics make membrane processes both energy-efficient and cost-effective compared with conventional separation methods, offering industries a practical route to improve sustainability and reduce overall operational costs [9–11]. Despite these strengths, several challenges persist, including membrane fouling, elevated energy demands in certain operations, limited membrane durability, and inherent trade-offs between selectivity and permeability, all of which continue to constrain widespread industrial adoption and optimization [7,9].

One of the most prevalent issues is membrane fouling, caused by the accumulation of organic and inorganic matter, microorganisms, and scaling agents on the membrane surface, which significantly reduces permeability and selectivity and increases cleaning requirements, downtime, and operational cost [12].

High energy demand remains another critical barrier, particularly in processes such as seawater RO and membrane distillation. Despite technological progress, seawater RO still requires approximately 2.5–3.1 kWh/m^3 of produced freshwater, while membrane distillation suffers from heat recovery inefficiencies and temperature polarization that restrict commercial deployment [13,14].

Furthermore, membrane systems often face the inherent selectivity–permeability trade-off, particularly in polymeric membranes. Emerging materials such as Mixed Matrix Membranes (MMM), incorporating fillers like Metal-Organic Frameworks (MOFs) or zeolites, have shown promise in overcoming this limitation by enhancing both properties simultaneously [15]. Mechanical weakness in hollow fiber membranes presents an additional constraint, though braid-reinforced designs have demonstrated improved durability under high-pressure operation [16].

Lastly, membrane durability and lifespan remain significant limitations, particularly in harsh operating environments involving aggressive chemicals or elevated pressures. Improving mechanical robustness and long-term stability through advanced materials, structural reinforcement, and optimized operating strategies is essential for lowering lifecycle costs and supporting sustainable membrane applications. Consequently, progress in membrane design, operational control, and performance optimization is critical for overcoming these limitations and enhancing membrane system reliability [17,18]. In this context, emerging tools such as Machine Learning (ML) models, Digital Shadows (DSs), and Digital Twins (DTs) offer promising pathways for predicting, monitoring, simulating, controlling, and optimizing membrane-based processes.

ML-based models, DS, and DT represent three distinct levels of digitalization in modern process systems. ML-based models are purely data-driven predictive tools that learn statistical patterns from historical or real-time datasets but do not maintain a continuously synchronized representation of the physical system nor influence its operation [19]. In membrane engineering, ML-based modeling contributes in three major areas: (1) predicting membrane properties to estimate key parameters such as permeability, selectivity, and structural characteristics; (2) using explainable artificial intelligence (AI) to uncover quantitative structure–performance relationships and provide deeper physical insight; and (3) enabling ML-guided design, optimization, and virtual screening of membranes to accelerate material discovery and process development [20–22]. DSs, in contrast, provide a unidirectional digital representation of a physical asset, in which real-time sensor data passively update the virtual model via a one-way connection, and no feedback is transmitted back to the physical process. As a result, DSs enable monitoring but not control. In membrane processes, DSs facilitate real-time performance tracking through data acquisition, enabling early fouling detection, operational-state visualization, anomaly identification, and condition-based maintenance strategies [23]. DTs represent the most advanced stage of digitalization, characterized by fully integrated, bidirectional, and dynamic data exchange between the physical and digital domains. This interaction enables real-time prediction, simulation, diagnostics, control, and optimization of the system [24,25]. In membrane technologies, DTs can serve as high-fidelity virtual replicas that predict multiphysical behavior, including mass transfer, hydrodynamics, fouling evolution, and chemical interactions under varying operating conditions. By extending the capabilities of both ML-based modeling and DSs, DTs support dynamic simulation, predictive control, fault diagnosis, optimization of operating conditions, and scenario testing to enhance membrane efficiency, durability, and sustainability across water treatment, wastewater management, gas separation, and H_2 production processes [24]. Fig. 2 presents the key differences between these three frameworks.

Although several review articles have examined the application of AI and ML in membrane materials and separation processes, most of these studies have focused on isolated aspects, such as data driven prediction of membrane properties, ML-assisted experimental screening, or general overviews of AI applications in water and gas separations. In contrast, the present review moves beyond an algorithm focused perspective by explicitly distinguishing and systematically integrating three related but conceptually different digital approaches, namely hybrid ML-based modeling, DSs, and DTs, within the context of membrane technologies. The essential role of computational fluid dynamics (CFD) within these frameworks is also highlighted, particularly its contributions to generating high-quality synthetic datasets, deepening mechanistic understanding, and supporting virtual experimentation for model development and validation.

More specifically, this review presents a unified and comparative framework that clarifies the functional roles, data needs, levels of physical fidelity, and decision-making scope of ML models, DSs, and DTs across various spatial and temporal scales, from membrane materials and modules to process-level operation, monitoring, and control, as shown in Fig. 2. A key and distinctive feature of this work is the explicit positioning of CFD as a physics-informed foundation within these digital frameworks. Instead of viewing it only as a simulation tool, this review highlights its role in generating high-quality synthetic datasets, ensuring mechanistic consistency in hybrid physics-based ML models, and aiding

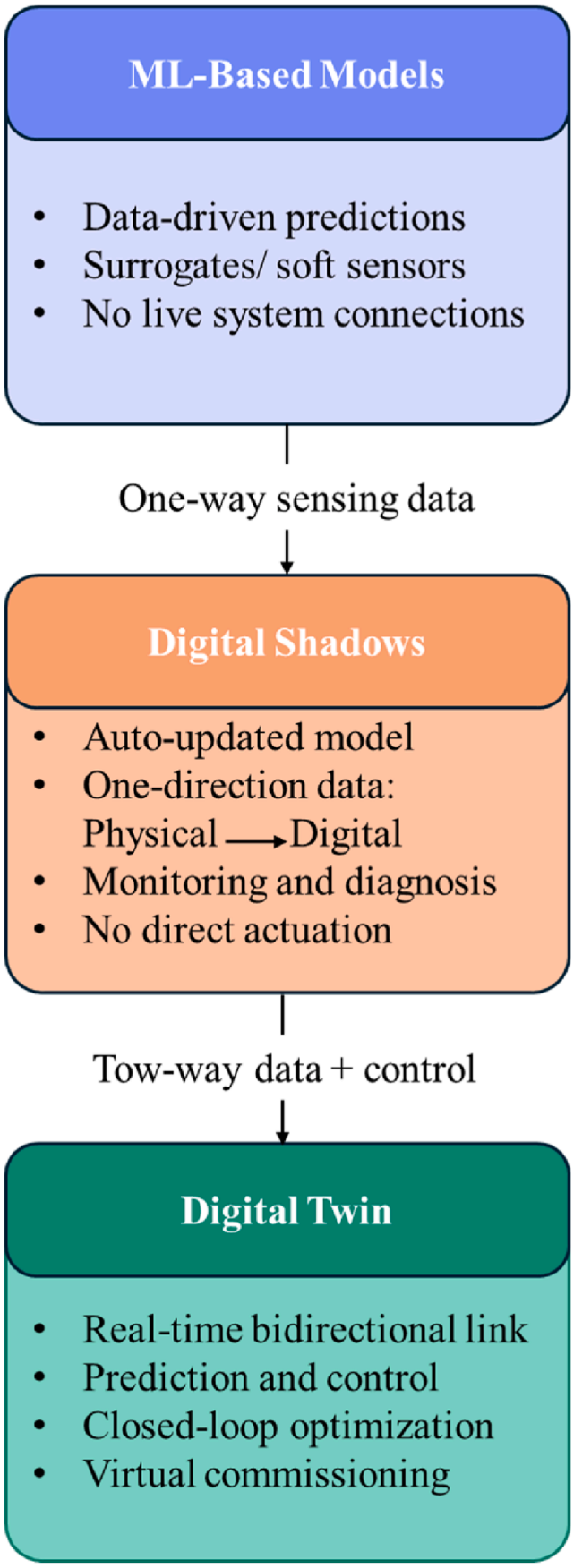


Fig. 2. Key differences between ML models, DS, and DT.

the development and validation of scalable digital representations of membrane systems.

By critically comparing existing approaches across model architecture, coupling strategy, computational cost, generalization performance, and relevance for industrial implementation, this review shifts the focus from proof-of-concept ML studies to operationally realistic and scalable digital solutions. It provides a structured roadmap for developing and deploying ML-enabled DSs and DTs in sustainable membrane technologies, including water purification, wastewater treatment, gas separation, MRs, and energy related applications. Additionally, key technological barriers, open research gaps, and future research directions are systematically discussed to facilitate the transition from laboratory-scale demonstrations to commercially viable digital membrane systems.

2. Fundamentals of ML-based models, digital shadow, and digital twin frameworks

2.1. Definition and key components

An ML algorithm is a computational method that performs a given task by exploiting input data rather than relying on explicitly defined, fixed rules. Instead of being hard coded, these algorithms adapt their internal structure autonomously through iterative exposure to data, thereby progressively improving their performance. During training, the model is gradually adjusted by providing it with input examples paired with their corresponding outputs. Through training, the algorithm learns to produce correct responses for the given examples and to generalize its predictions to new data. This ability to generalize is the key part of the "learning'' mechanism in ML. Importantly, training doesn't only take place at the start; many advanced algorithms are capable of continuous or "lifelong'' learning as new data becomes available [26]. A wide range of ML algorithms has been created to address the diverse types of data and problem structures faced in practice. The algorithms differ substantially in both how they represent these candidate models (for example, as decision trees, mathematical functions, or even full programming instructions) and in the strategies they use to navigate the search space. These strategies may include optimization techniques with formal convergence properties or evolutionary approaches that iteratively assess and refine generations of randomly altered model candidates [27].

ML-based models are inherently data-driven, typically relying on historical datasets obtained from experiments, simulations, or physical system modeling rather than on real-time operational data. Consequently, these models often operate offline to provide predictive insights. Depending on the data source, ML models can generally be categorized as experimental-based or hybrid. Fig. 3 schematically presents the architecture of an ML-based modeling system.

Experimental ML methods, while powerful, can face several challenges, including overfitting or underfitting due to limited or noisy data, restricted generalizability, and reduced accuracy in long-term forecasting. Hybrid ML approaches have emerged as an effective solution to these limitations by integrating data-driven learning with mechanistic models, including physics-based or model-based components (e.g., CFD simulations, process models). This combined method captures a wider range of system behaviors and takes advantage of both approaches' strengths, often leading to better predictive performance [28,29]. Mechanistic components retain essential process knowledge and can extend it beyond the calibration domain. In contrast, data-driven elements can identify complex, nonlinear relationships that first-principles models may not represent well [30,31] In many cases, hybrid systems include a dedicated prediction module and a secondary optimization module that systematically refines the predictive model to enhance accuracy. As a result, hybrid architectures offer greater flexibility and capability compared with single-method approaches, contributing to their rapid growth and widespread adoption in recent years [32].

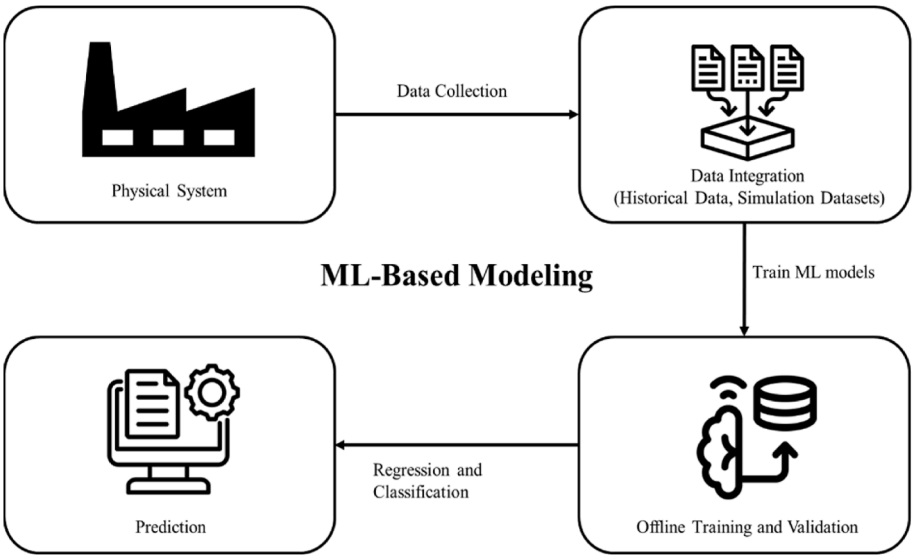


Fig. 3. Architecture of the ML-based modeling.

ML-based modeling is increasingly important in membrane science and engineering. It supports the development of advanced membrane materials, enhances process optimization, and facilitates system design and integration. These approaches are especially valuable for addressing the inherent nonlinearities and strong parameter interactions characteristic of membrane processes, which can make traditional modeling computationally expensive and time-consuming. Moreover, increasing environmental and societal demands underscore the need for efficient, data-driven methodologies that improve membrane performance while meeting sustainability objectives [33].

The DT is best described as a high-fidelity, dynamic virtual representation of a physical object, system, or process that is continuously updated with real-time data to reflect the current, past, or predicted state of the physical counterpart. The foundational concept of DT was introduced by Michael Grieves in 2003 and has since evolved into a pivotal technology within Industry 4.0 frameworks, enabling data-driven decision-making, optimization, and lifecycle management across diverse sectors [34]. Since the initial introduction of the concept, researchers have defined DT in various ways, depending on the field of application. It has been described as a virtual or digital model, a layout, a counterpart, a doppelganger, a clone, a footprint, a software analogue, a representation, an information construct, or a simulation of its physical entity. The earliest definitions were primarily shaped by the aerospace and aeronautics sectors, often using terms such as "aircraft," "vehicle," or "airframe." As the concept expanded into manufacturing and later into other industries, these terms were gradually replaced with broader descriptors such as "system," "machine," "product," "object," "entity," "asset," "device," and "process." More recently, the scope of DT has extended beyond nonliving or industrial objects to encompass complex biological systems, such as humans and trees. To avoid the variations in terminology, a generalized definition of DT that can be applied across disciplines and industries:

"A DT is a dynamic, self-evolving digital or virtual model or simulation of a real-world subject or object (whether a part, machine, process, or even a living being) that reflects the exact state of its physical counterpart at any given time through real-time and historical data exchange. Importantly, the relationship is bidirectional: the DT mirrors changes in the physical twin, and modifications in the DT are also reflected in the physical system [25]." A DT system, as shown in Fig. 4b, is comprised of four key components:

1. **The Physical System:** This refers to the real-world asset or process being modeled. It may include machines, membrane modules, chemical reactors, or entire treatment plants. Sensors and actuators embedded in the physical system collect operational data.
2. **The Virtual Model:** This is a computational replica of the physical system, which can simulate behaviors, predict failures, and evaluate different operational scenarios. The virtual model can range from simple input-output mappings to complex multiphysics or AI-driven simulations [35].
3. **Data Integration Layer:** This component ensures real-time synchronization between the physical and digital entities. It involves continuous acquisition, filtering, and contextualization of sensor data using Internet of Things (IoT) technologies, which are then used to inform and calibrate the virtual model [36].
4. **A Real-Time Feedback Loop:** The defining feature of a DT, this loop enables bidirectional communication: sensor data updates the digital model. In contrast, the model's insights can trigger actions in the physical system. This loop supports advanced functions such as predictive maintenance, adaptive control, and real-time process optimization [37].

On the other hand, a DS is a virtual model that passively reflects certain aspects of a physical system through a one-way data flow. This provides a practical option for organizations that do not have the infrastructure to build a fully integrated, multi-perspective DT. Instead of creating a new, combined database, the DS approach collects and organizes specific information taken from existing, often separate data sources. This enables real-time analytics when direct connections are blocked due to inconsistencies or limited transparency. Unlike a DT, which requires continuous bidirectional communication between the physical and virtual entities, the DS is restricted to unidirectional data transmission, allowing the virtual representation to update based on physical system states without influencing or controlling the physical system. As illustrated in Fig. 4a, the DS provides a one-way informational projection of the physical entity, enabling monitoring and analysis, but it falls short of the interactive capabilities that characterize a true DT (Fig. 4b). Despite its reduced complexity compared to DT, the DS concept has become increasingly relevant as an intermediate, scalable solution for data-driven decision support in industrial and engineering domains [38,39].

However, despite their increasing use in membrane systems, a major challenge remains the lack of standardized validation protocols. Current practices often rely on case-specific criteria, such as prediction error, flux deviation, or fouling-detection accuracy, without unified methods to evaluate digital fidelity, robustness under disturbances, or scalability.

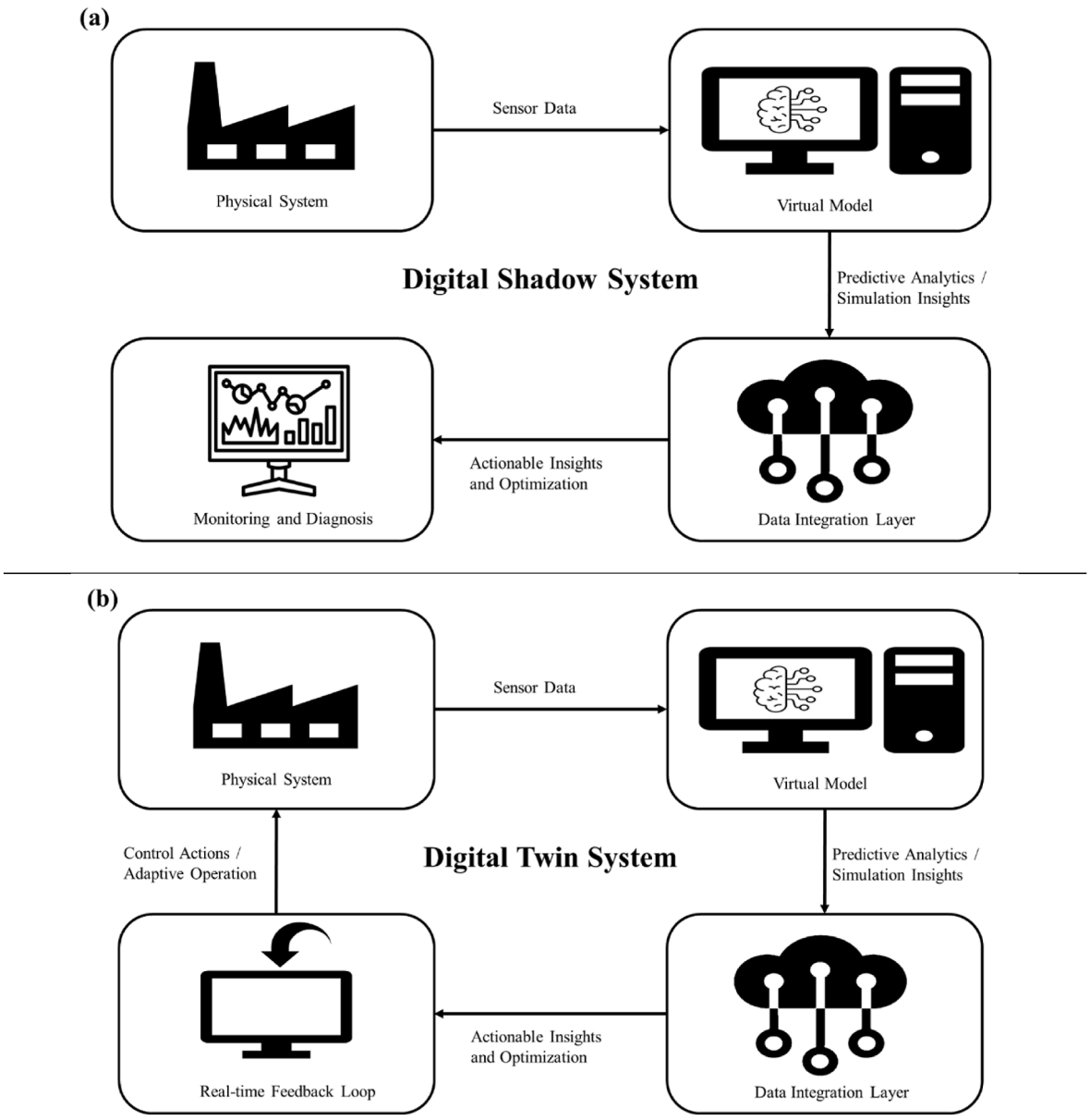


Fig. 4. Architecture of the (a) DS and (b) DT systems.

Developing domain-specific validation standards, including multi-level performance metrics (module, system, lifecycle), uncertainty quantification, stress testing under real-world conditions, and standardized data schemas, is a vital research priority. Solving this issue would greatly improve reproducibility, benchmarking, and industrial use of DS and DT frameworks in membrane technologies.

DS and DT frameworks are emerging as powerful enablers for advancing membrane-based separation systems, offering new capabilities for real-time monitoring, predictive optimization, and lifecycle management. DS provides a one-way virtual representation of membrane modules or filtration units, allowing operators to track performance indicators, fouling behavior, and process deviations using integrated sensor data without requiring a full cyber-physical infrastructure. In contrast, DT extends these capabilities by establishing bidirectional communication between physical and virtual entities, enabling dynamic simulation, adaptive control, and optimization of membrane performance under varying operational conditions.

These frameworks have been shown to improve process stability, energy efficiency, predictive maintenance, and operational decision-making in filtration and water treatment systems. As membrane systems become more complex and data-rich, DS and DT approaches, supported by IoT, improved sensing, and ML-enhanced modeling, are expected to play a central role in future smart membrane plants and Industry 4.0-aligned process intensification strategies [40–44].

2.2. *Types of applicable models*

In engineering and membrane processes, ML models are trained on experimental datasets, industrial sensor data, or high-fidelity numerical simulations such as CFD-generated synthetic data. Their role includes predicting performance indicators, generating virtual sensors, revealing structure–performance relationships through feature analysis, and accelerating process optimization. ML-based models are commonly classified into two categories based on the data source they use: (1) Experimental-based ML models, and (2) hybrid models that combine physical and data-driven elements [45,46].

DT and DS models can be applied to represent entities ranging from single components to complex, large-scale systems. While both aim to

create a virtual representation of a physical object or process, they differ in their degree of integration and interaction with the physical counterpart. In engineering and membrane systems, these virtual representations are typically organized at three levels—process, equipment (asset/component), and system—and both DS and DT can be instantiated at each level to support monitoring, simulation, while DT also supports control [23,38,47].

At the process level, DS and DT frameworks capture the dynamic behavior of an entire process line or treatment train. They integrate data streams from multiple units into a unified virtual environment, enabling real-time monitoring, fouling and performance prediction, what-if analysis, and process optimization. In membrane technologies, process-level models support proactive control strategies, energy-efficient operation, and better decision-making across interconnected stages of treatment [23,25,38,47–50].

At the equipment or asset level, DS and DT models focus on individual components such as pumps, valves, or membrane modules. These representations track component-specific performance, assess degradation mechanisms, and provide high-resolution diagnostics that support condition-based maintenance. By combining sensor data with virtual simulations of component behavior, equipment-level models help extend asset lifetimes, minimize downtime, and reduce maintenance costs while providing reliable insights into the health of critical membrane units [50,51].

At the system level, DS and DT models represent multiple assets operating together within a larger process environment, capturing interactions between units such as integrated membrane bioreactors (MBRs), desalination trains, or full-scale water treatment systems. These models facilitate lifecycle assessment, scenario evaluation, and optimization of system-wide efficiency under varying operating conditions. System-level DS and DT frameworks allow operators to evaluate coordinated responses, benchmark performance, and support strategic planning across complex, multi-unit membrane infrastructures [23,38,52,53].

2.3. Technological enablers

Hybrid models, DSs, and DTs share a common technological foundation. However, each needs different supporting technologies based on its level of digitalization. Hybrid models rely on high-quality datasets, advanced numerical simulations, and modern computational infrastructures as their primary enablers. Thus, CFD, high-performance computing, GPU/TPU acceleration, and data pre-processing pipelines form core enablers of the hybrid modeling ecosystem, together with learning frameworks such as Scikit-learn, TensorFlow, and PyTorch. These tools allow Hybrid models to capture nonlinear process–structure–performance relationships, support soft sensing, and develop surrogate models capable of approximating complex multiphysics behaviour with reduced computational cost [54–56].

DSs rely on the same computing and modeling tools as ML models but extend them through real-time data collection and system connectivity. A DS connects the physical system to a virtual model through a one-directional data flow. As a result, its performance relies heavily on IoT-enabled sensors, online analyzers, communication protocols, and continuous data streaming systems. CFD and other numerical models may also act as DS enablers when used as interpretive or predictive layers that process incoming sensor data, allowing the DS to contextualize real-time behaviour against high-fidelity virtual physics. Edge computing, time-series databases, and real-time analytics frameworks help the DS remain synchronized with the physical system. This allows for tasks like monitoring, soft sensing, diagnostics, anomaly detection, and performance tracking in MRs and chemical processes [57–59].

DTs represent the most advanced stage of digitalization and require the most comprehensive and tightly integrated technological ecosystem. DTs combine the enablers of both ML and DS with bi-directional communication and closed-loop control. Beyond real-time data acquisition, DTs can also integrate high-fidelity physics models (CFD, finite-element, or multiphysics solvers), reduced-order models, and hybrid physics–ML surrogates with industrial control systems to maintain continuous alignment with the physical system [40,59,60]. In membrane systems, CFD enables high-fidelity simulations of flow patterns, pressure dynamics, species transport, and concentration polarization. When continuously updated using operational data, CFD models form the virtual physics core of a DT, allowing it to predict system responses under varying conditions and evaluate operational scenarios in real time [61–63]. Such integration supports real-time prediction of phenomena such as concentration polarization and enhances flow distribution in spiral-wound or hollow-fiber membranes.

IoT and sensor technologies are also essential enablers, providing the real-time data needed to construct, update, and validate both DS and DT systems. Sensors embedded within membrane modules and process units measure key variables, including temperature, pressure, flow rate, and chemical composition. These measurements, transmitted through IoT networks, allow the virtual model to remain synchronized with the physical system and support predictive maintenance, fault detection, and adaptive control [35,64].

As the scale and frequency of sensor outputs grow, big data analytics become indispensable. Industrial processes generate massive, heterogeneous datasets that traditional methods cannot efficiently process. Big data analytics extract meaningful patterns from these datasets, enabling early fault identification, intelligent diagnostics, and optimized control strategies.

To handle these computational demands, cloud computing provides scalable platforms for data storage, model execution, and collaborative operation. Cloud-based architectures support DTs by allowing continuous integration of real-time sensor data, historical databases, ML training pipelines, and high-performance simulation tasks. When combined with big data analytics and AI, cloud infrastructures enable DTs to perform predictive maintenance, optimize resource allocation, plan operational strategies, evaluate dynamic responses, and make autonomous data-driven decisions [49,64]. Through the synergy of these technological enablers—AI and ML, CFD, IoT and sensors, cloud computing, and big data analytics—digital twins (DTs) transform from conceptual frameworks into fully operational, intelligent, and adaptive systems. The overall relationship among DTs, CFD, AI/ML, cloud and big data, and IoT/sensors is illustrated in Fig. 5. Furthermore, Fig. 6 highlights the difference between a DS and a DT.

3. Applications of ML-based models in membrane technologies

Recent advances in AI and ML have created new opportunities to accelerate innovation in membrane-based separation technologies. Over the past twenty years, data-driven approaches, including deep Learning (DL), neural networks, and hybrid ML-mechanistic models, have increasingly supported traditional physical modeling. They help address ongoing challenges, including fouling prediction, flux optimization, and energy-efficient process operation.

To explore the evolution of AI-enabled membrane research, a keyword co-occurrence analysis was conducted on Scopus-indexed publications from 2000 to 2025, covering titles, abstracts, and author keywords (retrieved on January 10, 2025). The results were analyzed using VOSviewer. As shown in Fig. 7, ML serves as the central research hub, strongly linked to DL, predictive modeling, and model interpretation. Key connections to membrane fouling, NF, UF, and flux prediction highlight the increasing use of AI tools to optimize membrane performance and operational reliability.

Despite the rapid growth of AI-, DS-, and DT-enabled applications in membrane technologies, a major challenge remains: the lack of standardized validation protocols. Current research often relies on case-specific performance metrics, such as prediction accuracy or flux deviation, without adopting unified methods to evaluate digital fidelity, robustness under disturbances, or scalability. Creating domain-specific

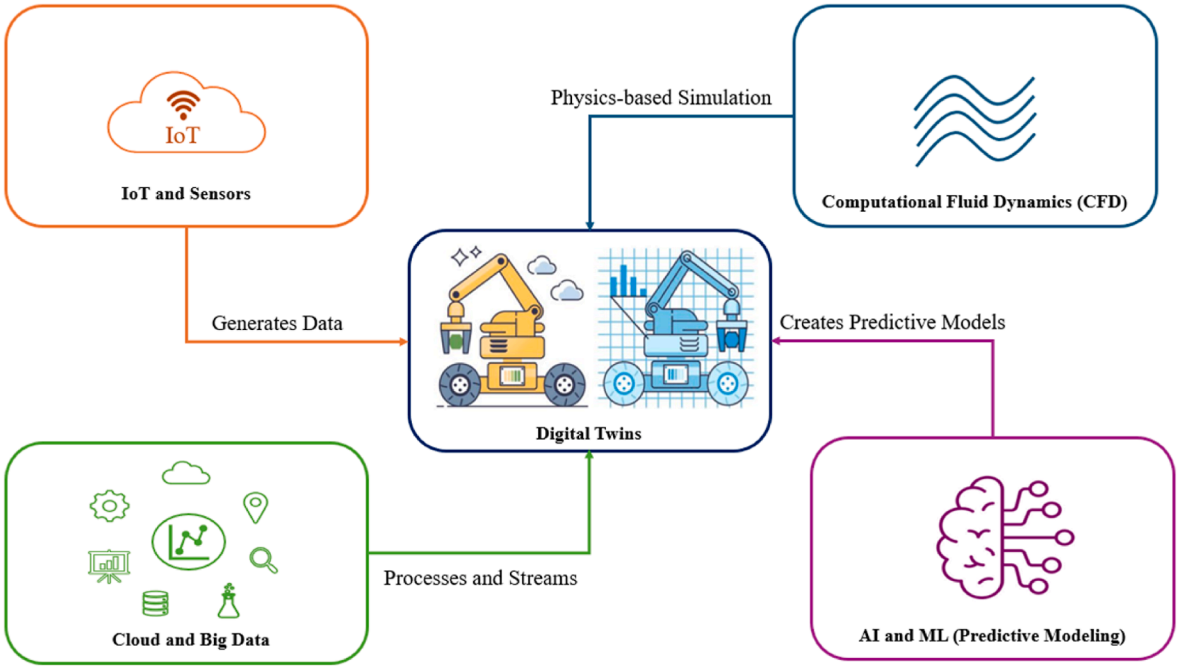


Fig. 5. Relationship between DTs, CFD, AI, ML, cloud, big data, IoT, and sensors.

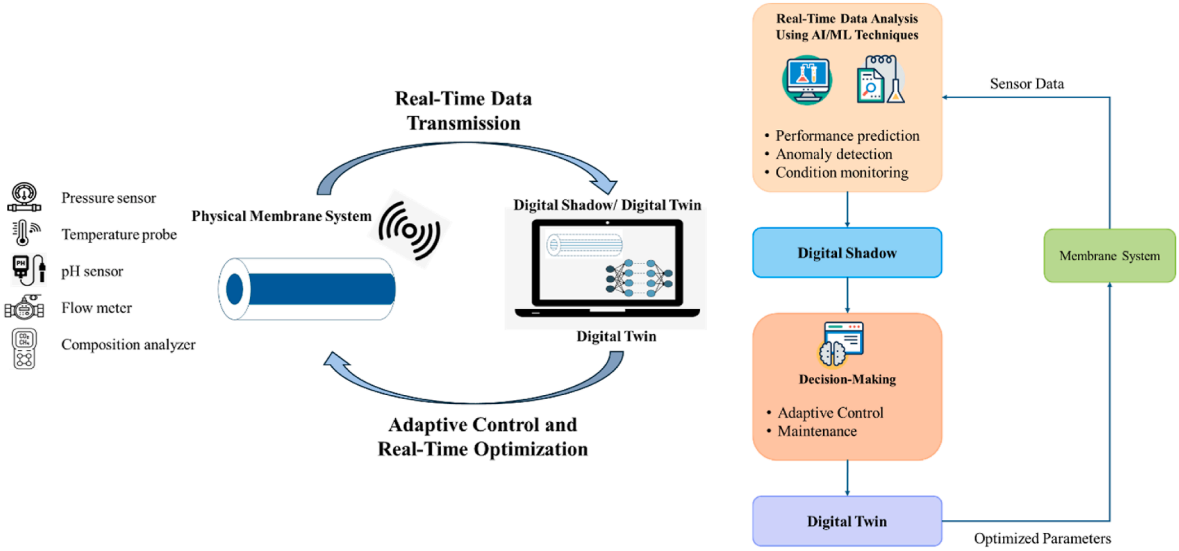


Fig. 6. Conceptual workflow linking ML-based analysis, DS monitoring, and DT decision support in membrane systems.

validation standards that include multi-level performance metrics, uncertainty quantification, stress testing under realistic conditions, and standardized data schemas is an essential research goal to improve reproducibility, benchmarking, and the industrial adoption of digital frameworks. Furthermore, the increasing appearance of terms such as DT, inverse design, and CO_2 and H_2 separation also reflects a shift toward intelligent, real-time membrane systems that support global decarbonization strategies. Collectively, these trends signal a transition from traditional performance modeling to integrated smart membrane platforms equipped with autonomous monitoring, control, and optimization capabilities.

Fig. 8 demonstrates how research themes at the intersection of AI and membrane technologies have evolved over time. Early studies (blue–green nodes, prior to 2015) focused primarily on fundamental modeling and mechanistic analysis, with keywords including mathematical model, molecular dynamics, and optical coherence tomography. These efforts marked the initial use of computational methods to investigate membrane structure, transport phenomena, and system behavior.

From around 2016 onward (green–yellow nodes), a pronounced shift toward data-driven methodologies is observed, with growing emphasis on ML, DL, artificial neural networks (ANNs), and performance prediction.

During this period, AI techniques increasingly targeted practical operational challenges, including fouling assessment, flux enhancement, and optimization of NF and UF processes—indicating a movement from theoretical modeling to performance-focused system improvement. In more recent years (yellow nodes, ~2020–2025), emerging topics such as DT, inverse design, CO_2 separation, H_2 separation, and CO2 capture and storage have gained prominence. These developments highlight the

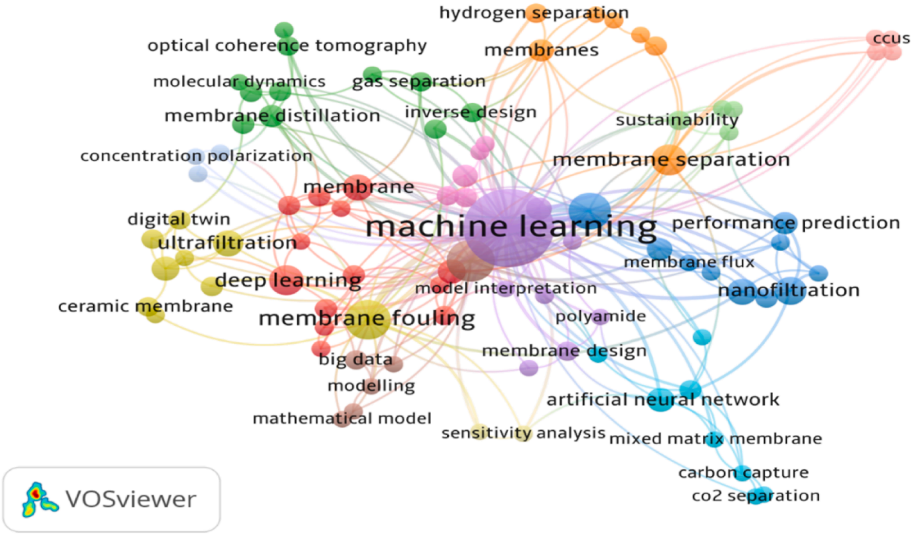


Fig. 7. Bibliometric co-occurrence map of research on AI, ML, and DT applications in membrane separation processes (2000–2025, Scopus; data retrieved Jan 10, 2025). Node size shows keyword frequency, link thickness indicates co-occurrence strength, and colors represent thematic clusters.

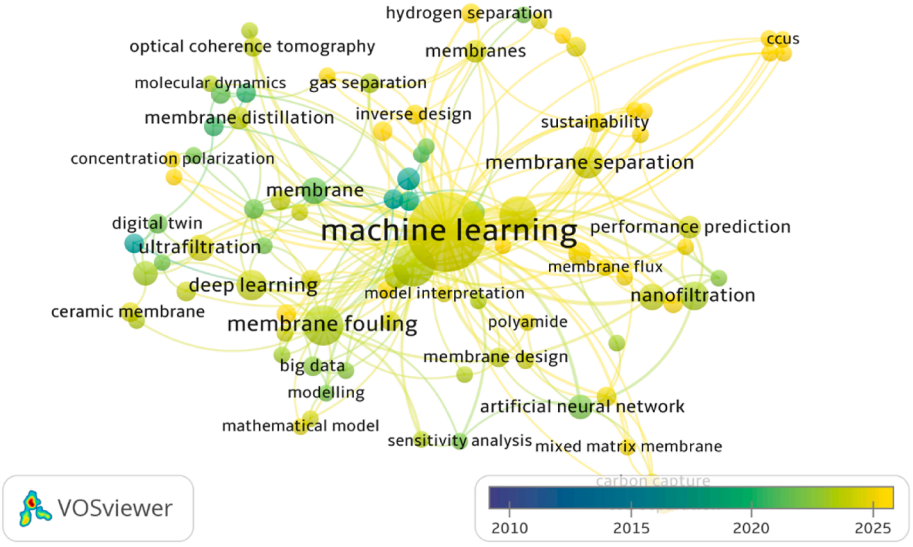


Fig. 8. Overlay visualization of keyword co-occurrence in AI-enabled membrane technology research (2000–2025, Scopus database), showing the evolution of research themes over time.

field's progression toward real-time intelligent membrane systems, autonomous control frameworks, and integration with broader clean energy and decarbonization technologies. Overall, the timeline underscores the rapid growth and diversification of AI-enabled membrane research, evolving from basic simulations to advanced, intelligent, and sustainable separation strategies.

Fig. 9 presents the annual publication trend for AI-driven membrane research between 2010 and 2026. Research activity remained modest prior to 2018 but increased steadily from 2019 onward, with a rapid acceleration after 2021.

The strong growth observed between 2022 and 2025 reflects heightened interest in applying ML, DL, and DT approaches to membrane systems—motivated by the need for improved predictive capabilities, advanced fouling mitigation strategies, and intelligent process optimization. The slight decline in 2026 results from incomplete indexing at the time of data extraction. Overall, the trend illustrates a rapidly expanding research domain aligned with global advancements in intelligent, adaptive, and sustainable separation technologies.

Fig. 10 shows the geographic distribution of research output related to AI-based membrane technologies. China and the United States lead in publication volume, followed by South Korea, Saudi Arabia, and Australia, indicating strong activity across major scientific and technological hubs. The involvement of additional countries—including India, the UAE, Canada, Germany, and Iran—highlights rising global engagement and significant opportunities for international collaboration in the development of intelligent membrane separation systems.

Despite the rapid growth and diversification of AI-enabled membrane research, a closer examination of the literature indicates that most reported studies remain limited to standalone ML models or offline simulations. Fully functional DSs, and especially industrial-grade DTs, characterized by continuous data integration, real-time updating, and closed-loop decision-making, remain scarce in membrane technologies. Accordingly, the following subsections critically review application-specific studies, distinguishing among ML models, DSs, and DTs based on their functional capabilities rather than their nominal classifications.

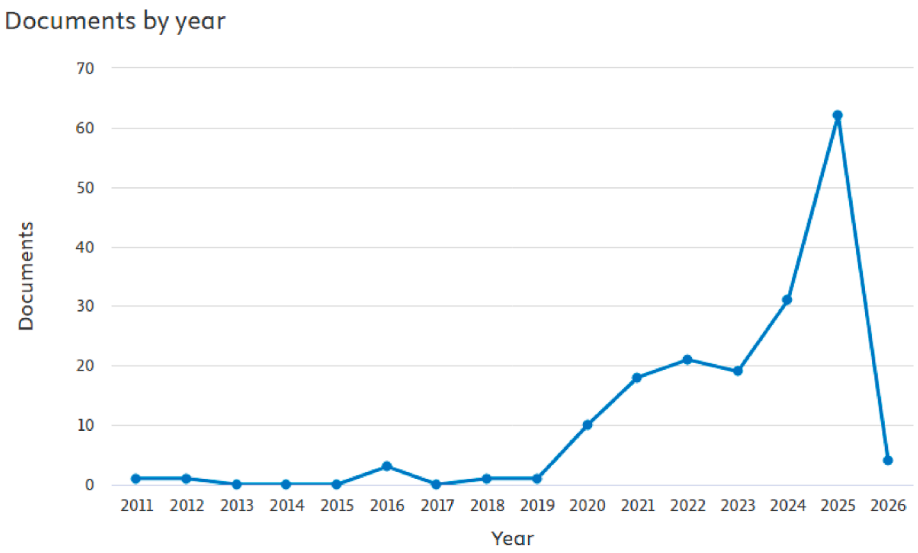


Fig. 9. Yearly scholarly output on AI and ML applications in membrane separation technology (2010–2026).

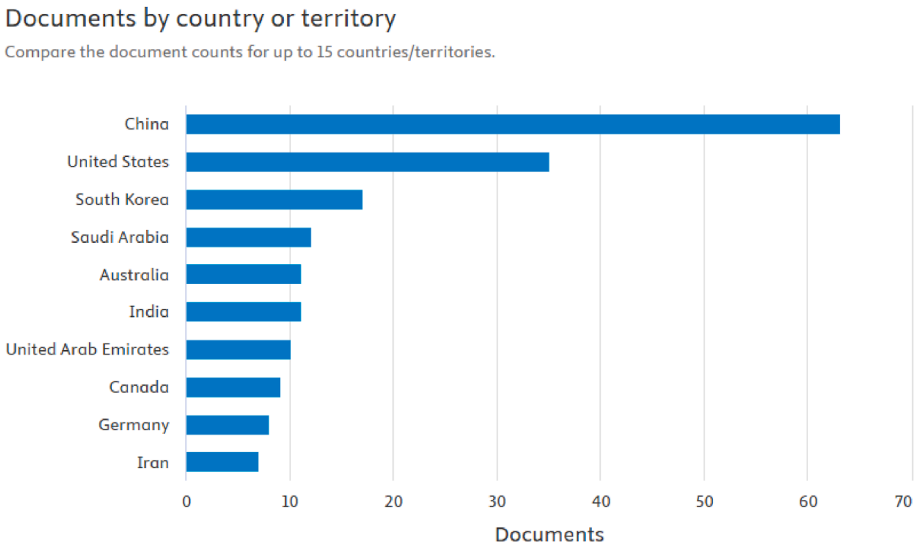


Fig. 10. Distribution of publications in AI-enabled membrane research by country.

3.1. Water/wastewater treatment via membrane technologies

AI-driven technologies, including hybrid modeling, DS, and DT, are gaining prominence in the water treatment and desalination sectors, offering powerful capabilities for managing large and complex water networks. As population growth drives rising water demand, these models provide developers, operators, and stakeholders with practical tools for designing, operating, and managing water infrastructure in the long term [65]. Water treatment infrastructure is a critical component of delivering clean and reliable water supplies to communities. The rate of clean water utilization plays a key role in sustaining water systems, as it directly impacts the stability and reliability of urban water resources. The integration of intelligent automation, supported by the IoT and ML, introduces a transformative pathway to improve efficiency and address pressing challenges in public water distribution.

Although several studies in water and wastewater treatment are described in the literature as DTs, a careful examination shows that many of them do not fully satisfy the strict DT definition outlined in Section 2. In practice, most reported implementations correspond to hybrid ML models or DSs that operate in a passive or semi-online mode, without full real-time synchronization and closed-loop control. In the following discussion, each study is therefore categorized based on its actual functional characteristics rather than the terminology originally used by the authors.

In this context, IoT devices strategically deployed across the water network enable the detection and continuous monitoring of essential parameters such as flow rate. DS and DT frameworks further strengthen this system by collecting real-time data from the filtration systems and transmitting it to the cloud. The acquired data undergoes preprocessing and mining to build a virtual representation of the physical system, thereby supporting intelligent decision-making in the distribution process.

While hybris models and DS architectures provide valuable insights into the behavior of water treatment processes, integrating DT frameworks with predictive analytics enables utilities to implement highly accurate predictive maintenance and fault-diagnosis strategies. This synergistic approach not only improves operational reliability and energy efficiency but also provides actionable insights to optimize treatment processes, ultimately contributing to more resilient, adaptive, and sustainable water management practices [66].

Membrane separation technologies are well-established methods for removing undesired particles from solvents such as water. These processes are typically classified according to membrane pore size: RO (<1 nm), NF (1–2 nm), UF (2–100 nm), and MF (100 nm–10 μm). A common challenge across all membrane processes is the decline in permeate flux over time. This reduction is primarily caused by particle accumulation on the membrane surface, either as deposited solids or as concentration polarization due to salt build-up, or by pore blockage, which may occur reversibly or irreversibly (the latter referred to as membrane fouling). Fouling not only necessitates costly pre-treatment steps but also leads to flux decline, requiring higher operating pressures and frequent chemical cleaning. Such interventions can damage membranes and accelerate their replacement, thereby increasing overall operational costs. To mitigate fouling and prolong membrane lifespan, systems are generally operated below a critical transmembrane pressure threshold [43,67]. To address these challenges, integrating hybrid models, DS, and DT offers significant advantages. By enabling predictive analysis and real-time monitoring of fouling dynamics and scalability in membrane operations, ML-, DS-, and DT-driven frameworks can capture the complex interactions among operating pressure, flux decline, membrane aging, and cleaning cycles. Such capabilities provide operators with actionable predictive insights to balance operational performance with long-term membrane durability. Consequently, DS- and DT-assisted monitoring not only enhances energy efficiency through optimized pressure regulation but also minimizes the frequency of chemical cleaning and reduces the risk of premature membrane replacement. Ultimately, the adoption of AI-driven digital technologies in membrane processes improves operational reliability, cost-effectiveness, and long-term sustainability in water treatment systems.

In line with these challenges and opportunities, recent studies have explored the application of AI-driven models in water treatment and desalination, focusing on their potential for intelligent fouling monitoring, scalability assessment, and optimization of energy consumption and cost. Membrane-based separation processes are widely used at an industrial scale and are increasingly considered suitable for automation. However, their complex interactions and inherent uncertainties limit the development of rigorous theoretical models. For this, Møller et al. [43] introduced a DT methodology termed Stochastic Greybox Modeling and Control (SGMC), specifically applied to water UF. Unlike other probabilistic DT approaches, SGMC employs stochastic differential equations within a physically intuitive framework to explicitly capture uncertainties and dynamically update model states. The study demonstrated that SGMC can optimize filtration control and minimize energy consumption under fixed water volumes, highlighting its adaptability and robustness in real-world membrane operations. This framework can be classified as a grey-box DT–like system, as it supports online state estimation and control optimization but does not implement a fully autonomous closed-loop, plant-wide feedback architecture.

Prado-Rubio and Huusom [68] developed a DS framework to support adaptive operation of dynamic UF in the oil recovery industry, a sector with high water demand and significant environmental impact. Using data from 18 pilot-scale experiments, their approach combined signal preprocessing with a recursive least squares autoregressive model to decouple applied and unknown disturbances. This enabled accurate prediction of transmembrane pressure and membrane flux, with forecasting errors below 10%, comparable to ML and hybrid models. Notably, the method also provided reliable forecasts of fouling rates, offering valuable insights into the onset of irreversible fouling, and enhancing the understanding of complex UF behavior under uncertain conditions. Functionally, this implementation aligns more closely with a DS, since it continuously updates a virtual representation using process data without directly actuating operational decisions.

Van Rooij et al. [69] developed a DT–based decision support system (DSS) for optimizing membrane restoration strategies in RO desalination plants. Their approach models wear and repair at the level of individual membrane elements, capturing biofouling dynamics driven by seasonal algal blooms. The DSS used differential pressure monitoring, biofouling-position effects, and particle filtering to estimate model parameters with high accuracy. This framework enabled operators to compare competing restoration policies with respect to cost, downtime, and risk, with predictions indicating 50–80% reductions in element replacement rates and substantial cost savings. The study demonstrated the potential of DSS to become a gold standard for long-term membrane asset management in desalination systems. Although referred to as a DT–based decision-support system, the proposed framework primarily operates as a semi-online DT focused on strategic maintenance planning rather than real-time closed-loop control. Cuba et al. [70] developed a DT of an RO desalination plant to optimize operation under variable electricity tariffs. The DT models, fitted and periodically updated with plant data, forecast permeate flow and energy consumption as a function of desalination pressure. By coupling these models with a genetic algorithm, the authors optimized hourly operating pressures to minimize electricity costs while meeting production targets. Case studies demonstrated that the DT-based strategy significantly reduced monthly operating costs compared with fixed-pressure operation, highlighting the economic advantages of DT-enabled energy optimization in desalination systems. This study represents one of the closest examples of a process-level DT in membrane desalination, integrating real-time plant data, predictive modeling, and operational optimization.

Furthermore, to mitigate membrane fouling during filtration, integrating membrane systems with complementary processes—particularly oxidation techniques such as ozonation—has proven effective. Consequently, recent research in water purification has also focused on the development of AI-driven approaches for membrane ozonation systems [71–73].

Additionally, several explicit DS implementations for real-time membrane monitoring have recently emerged. Zadkarami et al. [74] developed a DS-based soft-sensor for dynamic UF, where online measurements of transmembrane pressure, flow rate, and backwash events continuously update an autoregressive model to forecast fouling rates with <10% error. Taylor et al. [75] reported a live DS soft sensor for single-pass tangential flow filtration (TFF) in bioprocessing, integrating mechanistic filtration equations with a Kalman-filter estimator to infer non-measured membrane resistance in real time. Heusel et al. [76] similarly implemented a hybrid DS-like metamodel for industrial dynamic cross-flow filtration, continuously updated with torque and pressure signals to provide near-real-time recommendations for permeate-flow setpoints. In drinking-water treatment, Gümüş et al. [77] developed a DS-style data-driven monitoring framework for hybrid adsorption–UF systems, using XGBoost surrogate models to forecast fouling, Dissolved organic carbon removal, and UV254 response based on online sensor inputs. Collectively, these studies demonstrate that DS frameworks for real-time fouling diagnosis and membrane-health tracking are emerging but remain limited to UF, RO pretreatment, and TFF systems, with almost no validated DS implementations for gas separation, CMRs, or high-temperature hydrogen-production membranes, highlighting a critical research gap.

Table 1 summarizes representative studies on hybrid modeling frameworks applied to membrane filtration and membrane ozonation processes in water treatment.

Collectively, research on hybrid modeling, DS, and DT architectures for membrane-based water treatment demonstrates their strong potential to enable accurate, cost-efficient system design, performance prediction, and operational monitoring. Fig. 11 illustrates the hybrid modeling, DS, and DT frameworks applied to membrane filtration processes.

Overall, although water and wastewater treatment are among the most advanced applications of AI-enabled membrane systems, fully industrial-grade DTs with continuous synchronization, standardized data interfaces, and closed-loop control remain limited to a few pilot and plant-scale demonstrations (see Fig. 12).

Table 1
Hybrid modeling studies on design, operation, and optimization of water treatment.

Process	Modeling Approach	ML Architecture	Hybridization	Generalization/Physical Consistency Approach	Ref.
Multi-stage RO	Aspen-based RO simulation + ML surrogate	Artificial neural network (ANN) (feed-forward, regression) and support vector machine (SVM) (RBF kernel) were used for stage-wise prediction of permeate quality and pressure behavior	Aspen Plus simulation generated input–output pairs (T, P, salinity → flux/recovery). The ML surrogate predicted stage performance and captured nonlinear effects not captured by Aspen.	Normalized datasets; train/val/test split; ANN reduced MSE significantly in the second stage; SVM suffered higher MSE due to kernel mismatch. No physical constraints; generalization limited to Aspen-generated domain.	[78]
Key Technical Insights (Advantages/Limitations): ANN achieved lower MSE and captured nonlinearities in brine concentration more effectively than SVM; however, reliance on synthetic data limited extrapolation to plant conditions. Useful for design-scenario exploration, but lacked fouling physics; therefore, not suitable for real-time control.					
UF membrane process	Dynamic grey-box hybrid model	ANN (single hidden layer) predicting fouling resistance derivative	Darcy's law governed flux; ANN predicted the time-derivative of resistance to capture nonlinear/time-variant fouling. CFD not used; relied on plant data.	Used normalized field data sampled every 3 s; physical consistency enforced through the Darcy equation. ANN structure prevented overfitting by modeling only the unknown fouling term.	[79]
Key Technical Insights (Advantages/Limitations): Strong for dynamic fouling prediction under backwash/back-shock cycles; hybridization ensured physical realism. Limitation: ANN training required sufficient variability in cleaning cycles; the model could have underperformed on unseen feed characteristics.					
UF membrane process	Practical hybrid model for industrial-scale UF	ANN (multilayer feed-forward) trained on turbidity, TMP, and filtration time	Cake filtration model gave structural resistance; ANN predicted specific cake resistance (α) and total suspended solids (TSS) when real lab measurements were unavailable.	Dataset from pilot plant; physical parameters bounded via Darcy's law; sensitivity analysis confirmed turbidity as dominant input, supported physical consistency.	[80]
Key Technical Insights (Advantages/Limitations): Accurately estimated fouling indicators using only online plant data; good generalization as turbidity strongly correlated with α. Limitation: correlation turbidity–TSS changed seasonally, and periodic retraining required.					
Membrane-based ozonation	CFD + ML surrogate model	Ensemble trees (Extra Trees, Random Forest, Gradient Boosting (GB), Adaptive Boosted Trees (ADT)), hyperparameter-tuned using Firefly Algorithm	CFD solved mass-transfer/reaction; 10,000 samples used to train ML surrogate C(r,z). ML surrogate provided real-time prediction of ozone distribution replacing expensive CFD.	Outlier removal via Mahalanobis distance; min–max normalization; cross-validated ensemble models; physical consistency indirectly preserved by dense CFD sampling.	[81]
Key Technical Insights (Advantages/Limitations): The ADT model achieved $R^2 \sim 0.999$ and the lowest max error. Advantages: very fast inference, strong nonlinear mapping. Limitation: purely data-driven; no explicit chemical reaction physics embedded, which may fail outside the CFD-training envelope.					
Membrane-based ozonation	CFD + ML comparison study	Convolutional Neural Network (CNN), Support Vector Regression (SVR), (RBF kernel), and Orthogonal Matching Pursuit (OMP) (linear sparse regression) with Glowworm Swarm Optimization hyperparameter tuning	CFD provided spatial C(r,z); ML predictions compared architectures. SVR captured smooth nonlinear surfaces; CNN captured local spatial correlations.	Subspace outlier detection; min–max scaling. No physics constraints; generalization limited to CFD sampling.	[82]
Key Technical Insights (Advantages/Limitations): SVR ($R^2 \sim 0.99698$) outperformed CNN ($R^2 \sim 0.98$) and OMP ($R^2 \sim 0.87$). CNN is computationally heavy for simple r–z grids. OMP failed to capture strong nonlinearity.					
Porous membrane contactor (ozonation)	CFD + boosted ML surrogate	AdaBoost-K nearest neighbor (KNN), AdaBoost-decision tree (DT), AdaBoost-Automatic relevance determination (ARD); hyperparameters tuned via Successive Halving; outlier removal via Isolation Forest	CFD simulation produced the ozone concentration C(r,z); boosted learners predict the concentration surface. Boosting leverages multiple weak learners to form an accurate surrogate.	Strict data cleaning (Isolation Forest); 75/25 train–test split; performance tested via R^2, RMSE, and max error.	[83]
Key Technical Insights (Advantages/Limitations): AdaBoost-KNN achieves $R^2 = 0.9992$, RMSE$\approx 1.5 \times 10^{-2}$, indicating the best model. Strength: robustness against irregular CFD gradients. Limitation: physically unconstrained and dependent on CFD mesh fidelity.					

These studies collectively emphasize fouling prediction accuracy, energy efficiency, and operational robustness as common evaluation criteria across different membrane configurations.

3.2. Membrane bioreactors (MBRs) in wastewater treatment

The integration of modeling, simulation, and digital threads with AI-driven, DS, and DT technologies is expected to significantly advance the planning, design, and management of wastewater treatment systems. As the connection between hybrid models, DSs, and DTs and innovative water management becomes increasingly evident, wastewater treatment plants are being driven to adapt and optimize their operational strategies in line with these emerging tools [84]. Among existing treatment technologies, MBRs—which couple physical separation with biological treatment—are recognized as particularly promising for wastewater treatment, water reuse, and nutrient recovery in decentralized systems [85]. Compared with conventional processes, MBRs offer clear advantages, including simplified flow configuration, consistent production of high-quality effluent, high volumetric loading capacity, reduced sludge generation, and stable biomass retention without settling issues. Nevertheless, membrane fouling and limited membrane lifespan remain critical barriers to their broader application, as these factors elevate operational costs through frequent cleaning and membrane replacement [86,87]. Membrane fouling remains a critical limitation in MBR operation, as it directly impacts membrane flux and longevity. Fouling results in reduced permeate flux, higher feed pressures, lower productivity, increased downtime, and higher operational costs due to more frequent cleaning, while also reducing the lifespan of the membrane module. Consequently, developing effective strategies to mitigate fouling represents the central challenge in advancing MBR applications [88]. To address this, extensive research has been conducted to elucidate fouling mechanisms and propose effective mitigation strategies. In this context, DT frameworks and AI-based techniques show strong potential for enabling predictive maintenance and optimizing process parameters,

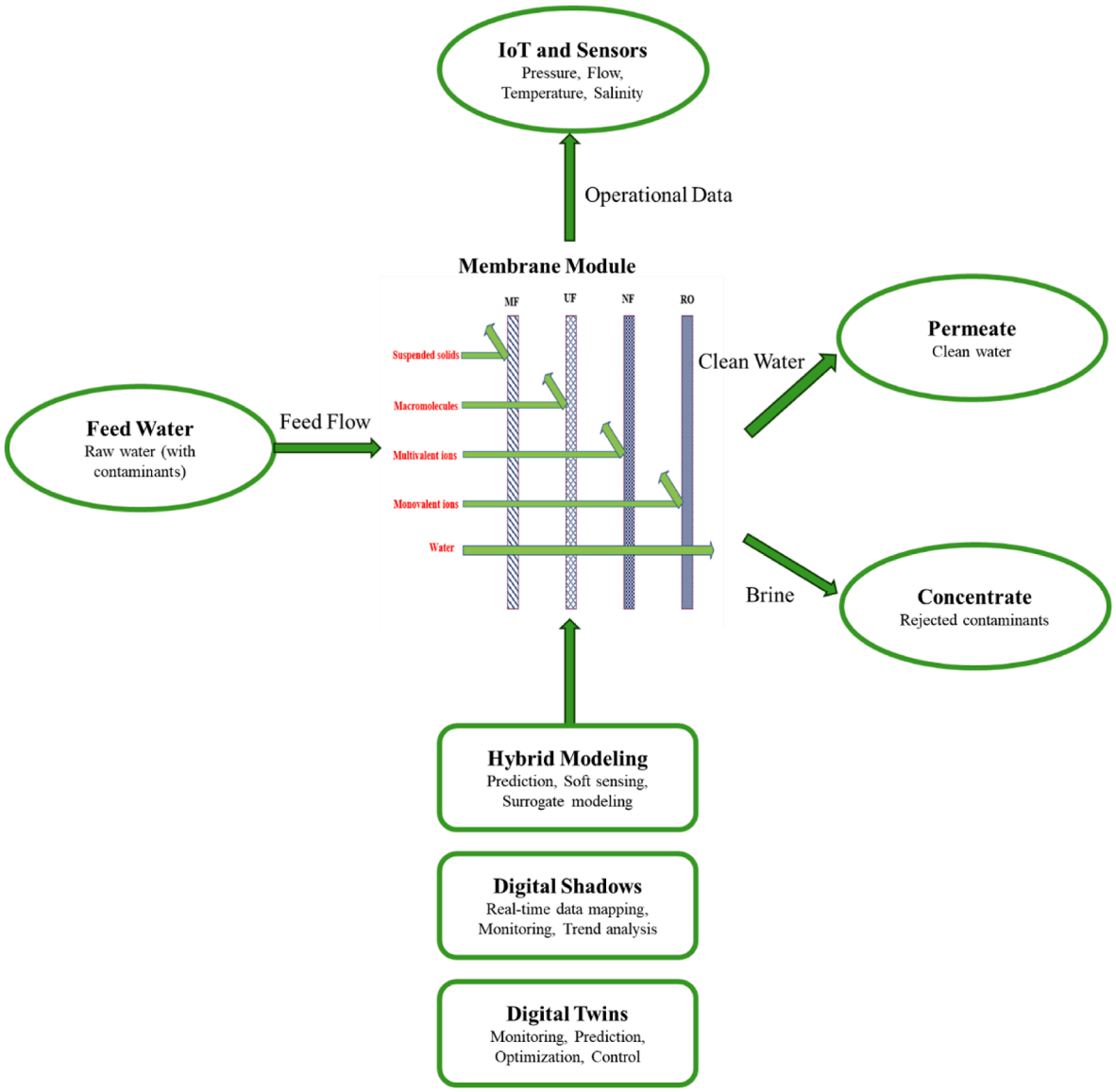


Fig. 11. Hybrid modeling, DS, and DT frameworks for membrane-based water treatment.

thereby supporting more cost-effective and sustainable operation. Moreover, these frameworks have also been applied to process control and fault detection in MBR systems [89]. Table 2 summarizes recent studies on hybrid modeling and DT applications for MBRs in wastewater treatment. Overall, these studies underscore the significant potential of DT and AI frameworks to improve fouling prediction, process optimization, and fault detection in MBR systems. However, after careful review of the available literature, a clear gap emerges in the application of DS frameworks to MBR wastewater treatment. Despite their suitability for real-time monitoring, anomaly detection, and dynamic performance assessment, DS approaches remain largely unexplored in this domain. Given that MBR systems generate extensive, high-frequency operational data, the integration of DS could offer substantial advantages in tracking system behavior, supporting early-warning mechanisms, and enhancing operational decision-making.

Furthermore, despite notable advancements in DT- and AI-enabled MBR research, challenges such as limited data availability, model interpretability, and integration with real-time plant operations continue to hinder large-scale implementation. Overcoming these limitations is essential to fully harness the capabilities of DT- and DS-enabled MBRs for sustainable, intelligent wastewater treatment.

3.3. Gas separation via membrane technologies

The growing reliance on fossil fuels has intensified climate change. This highlights the importance of renewable energy as a main approach to reduce global warming. Emerging digital technologies, including hybrid models, DSs and DTs, offer powerful tools for optimizing energy generation, storage, and utilization, thereby enhancing the efficiency, flexibility, and resilience of modern energy systems. While physical and mathematical models have long been used to simulate processes such as renewable energy generation and storage, their reliance on parameters and assumptions can limit accuracy. In contrast, AI, ML, DL, and more recently DSs and DTs, have proven highly effective in modeling and optimizing a wide range of processes, including H_2 production, biodiesel, and syngas. Within the energy sector, digitalization technologies not only enhance system performance, efficiency, and profitability but also help reduce environmental impacts. Fig. 13 illustrates the hierarchical digitalization pathway for membrane-based gas separation processes, progressing from CFD and mechanistic modeling to AI/ML, hybrid frameworks, Ds, and ultimately DTs.

Ongoing advancements in AI, ML, DS, DT, and intelligent control technologies are expected to further accelerate digitalization within the

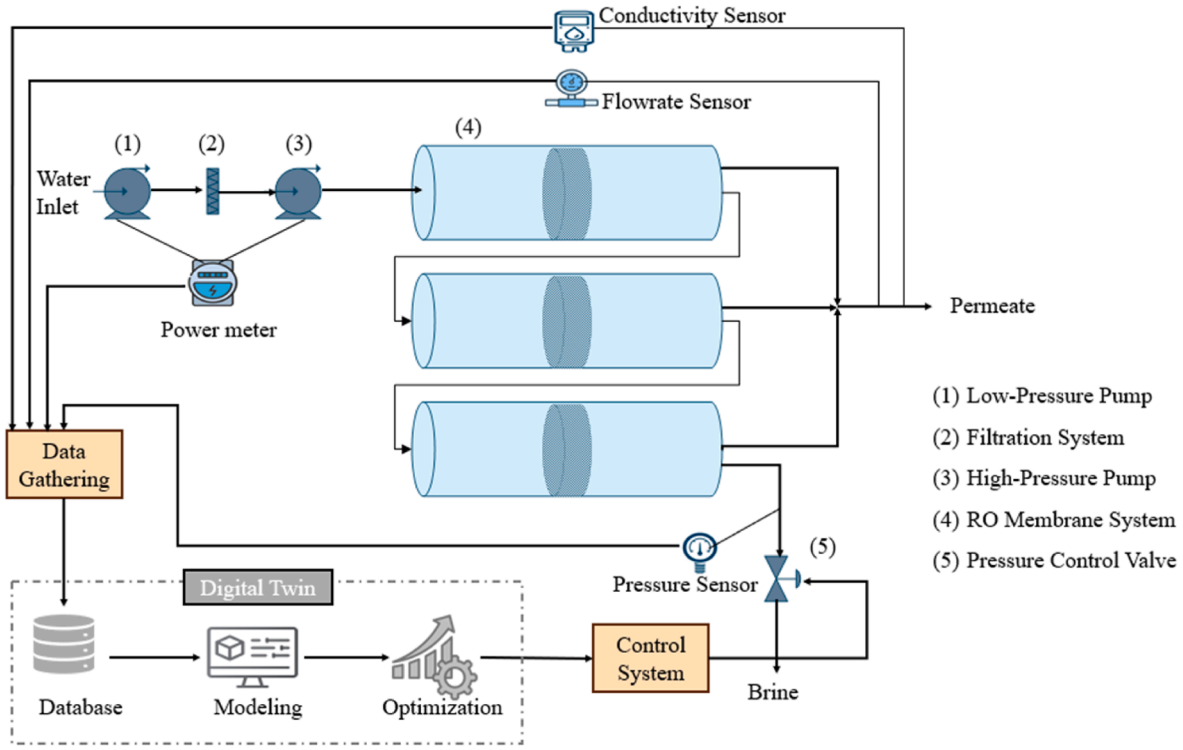


Fig. 12. Closed-loop DT representation of a RO desalination plant developed by Cuba et al. [70].

Table 2
DT and hybrid modeling studies on design, operation, and optimization of MBR wastewater treatment.

System	Goals	Feedstock	Approach	Modeling	Ref.
Full-scale MBR	Operation/ Optimization	Food-processing industrial wastewater	Hybrid modeling	Predictive fouling framework using CatBoost ($R^2 = 0.8374$), moving-average temporal smoothing, COD-removal refinement, and XAI identification of dominant drivers (F/M ratio, MLSS). Supports proactive fouling management and provides interpretable decision support; requires validation across diverse datasets.	[90]
Submerged MBR (full-scale fouling forecasting)	Operation/ Optimization	Municipal WWTP	Hybrid modeling	The integration of high-quality data collected from full-scale WWTPs and the application of advanced data-cleaning and model-training techniques have led to the establishment of an ML framework (MBR-Net) that enables real-time prediction of membrane fouling, expressed in terms of permeability.	[91]
High-salinity MBR (nutrient removal)	Design/Operation/ Optimization	Industrial/high-salinity WW	Hybrid modeling	Mechanistic model of Activated Sludge Model 3 outputs + interpretable ML (CatBoost/SHAP) for predicting biodegradability and treatment efficiency	[92]
Centralized MBR plants (case-study DT)	Design/Operation/ Optimization	Municipal WWTPs	DT	DT for proactive control and soft sensing	[93]
Baffled MBR	Design/Operation/ Optimization	Industrial wastewater	DT	DT to simulate biological performance, effluent quality, and process kinetics and "virtual piloting" to reduce real-life pilot testing and support optimal design modifications	[94]

energy sector, enhancing operational efficiency and supporting the transition toward greener and more sustainable energy systems [95].

Membrane processes are widely employed in gas separation, as illustrated in Fig. 14, due to their energy efficiency, compact design, scalability, and ability to deliver high selectivity and permeability. To support industrial deployment, membranes are synthesized from a range of materials and configured in geometries such as tubular or planar, which are then integrated into specialized modules. These modules are designed to minimize the required membrane surface area and enhance mass transfer, thereby enabling the feasibility of large-scale gas separation. For practical design and optimization of such systems, accurate modeling and simulation frameworks are essential. These frameworks must capture the influence of membrane material properties, geometry, and operating conditions on overall performance, while also accounting for non-idealities such as flow maldistribution—a standard limitation in industrial-scale membrane units [96]. Benefiting from hybrid models, DSs, and DTs in membrane-based gas separation processes are still emerging areas, and research to date remains limited. Nevertheless, existing studies demonstrate the effectiveness of these models for prediction and optimization tasks in membrane-based gas separation systems. With the rapid advancement of AI, a growing number of contributions can be expected in the coming years. Recent advances highlight the application of hybrid models, DS, and DT to enhance membrane efficiency in gas separation processes involving CO_2, H_2, O_2, and N_2. These AI-driven technologies have also been applied to optimize MR systems for biodiesel and biogas purification. Across these applications, data-driven approaches have demonstrated significant gains in efficiency and selectivity, enabling the development of specialized membranes with broad potential. Such integration offers versatile pathways for tackling global sustainability challenges, supporting carbon capture, industrial gas purification, and the reduction of CO_2 emissions [97,98].

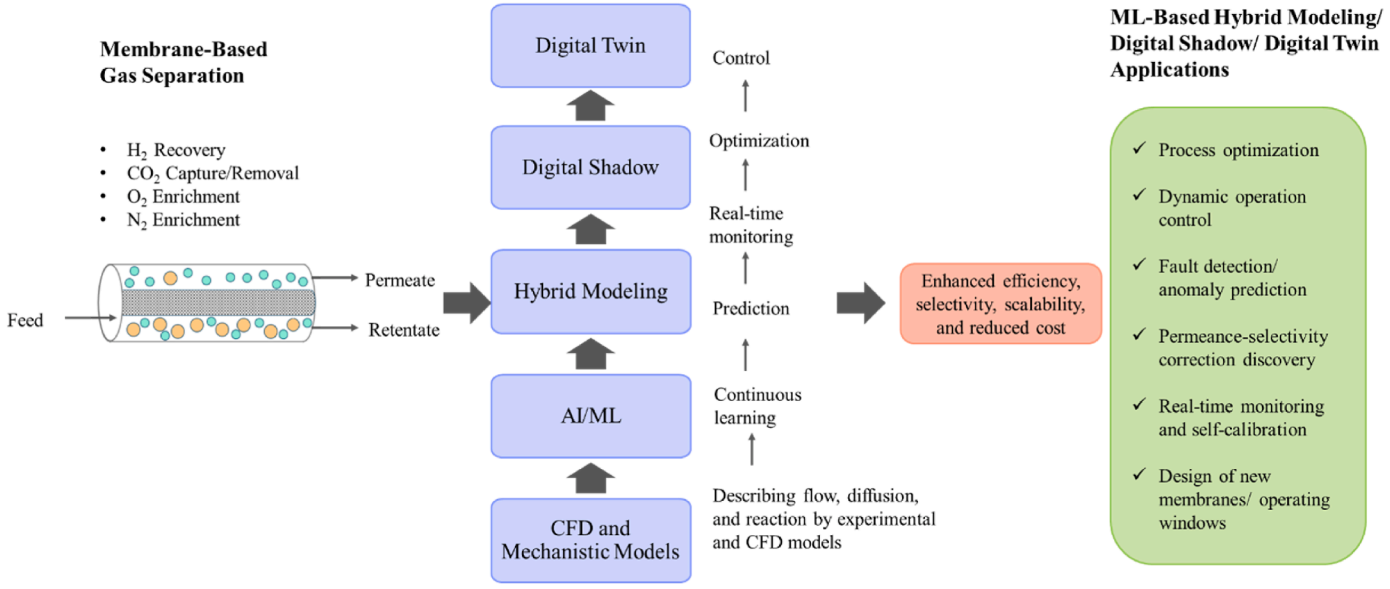


Fig. 13. Digitalization framework for membrane gas separation processes, illustrating the roles of mechanistic/CFD models, AI/ML approaches, DS, and DT within modern digital membrane engineering.

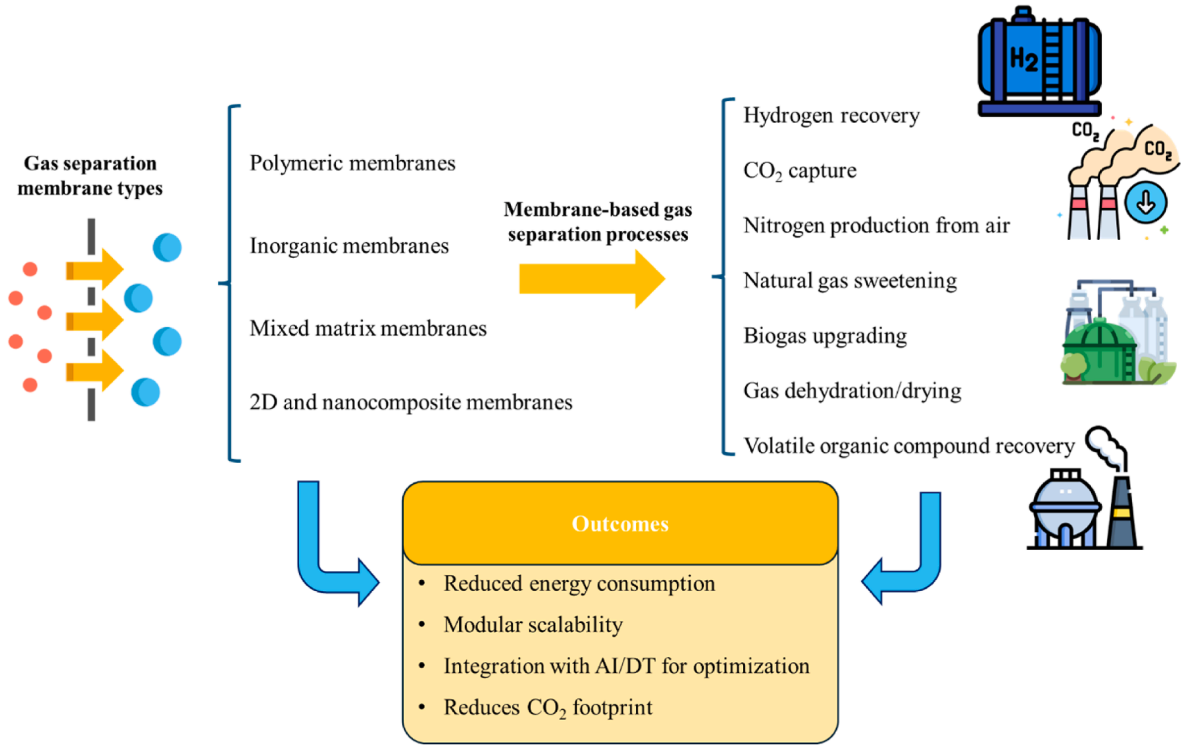


Fig. 14. Potential applications and benefits of membrane technology for gas separation.

3.3.1. *Carbon capture process*

Despite continuous progress in renewable energy technologies, the rapid growth in global energy demand has sustained the dominance of fossil fuels, which still account for nearly 80% of primary energy consumption. Consequently, global CO_2 emissions continue to rise. Among all sources, coal-fired power plants are the largest point-source emitters, with thousands of units currently operating worldwide. Reducing their emissions is therefore a critical priority as nations pursue decarbonization strategies. Carbon capture and storage (CCS) remains the only viable approach for mitigating CO_2 emissions from coal-based power generation while maintaining plant operation. Available CO_2 separation methods can be further classified into physical methods, such as absorption, adsorption, and membrane processes, and chemical methods, including solvent absorption, catalytic hydrogenation, and enzymatic extraction. Among these, membrane-based gas separation has gained attention as a promising CO_2 capture technology due to its operational simplicity, compact design, ease of scaling, environmental friendliness (no solvent emissions), and ability to operate exclusively with electricity. Membranes are also highly compatible with hybrid systems, allowing for seamless integration with other separation techniques.

However, membrane processes face challenges such as equilibrium limitations and constraints on the driving force for separation, which can limit large-scale applications economically. To overcome these barriers, continuous monitoring and precise control of operating parameters are crucial for achieving maximum efficiency [99].

DS, DT, and AI-driven technologies offer a promising pathway to address these challenges by enhancing efficiency, improving flexibility, and reducing maintenance costs. Despite their potential, the deployment of these technologies in power plants remains in its infancy, with limited applications and no reported implementations in gasification systems [100]. While research on hybrid framework applications for CO_2 capture remains relatively limited, the body of literature is steadily growing [100]. To our knowledge, no studies have reported the full implementation of DS and DT specifically for CO_2 capture membranes. This highlights a significant research gap and an opportunity for future work. The absence of DS and DT in membrane-based CO_2 capture processes is due to specific challenges in this area relative to generic process plants. First, the post-combustion and pre-combustion capture processes are highly multiscale. Material-level transport in membranes, sorbents, or ionic liquids must connect with complex plant-level heat and mass-transfer networks and, in some cases, subsurface storage models. This connection makes it difficult to calibrate and maintain accurate DS and DT in real time [48,101]. Second, industrial capture units are often retrofits on older plants with different control systems and limited tools for membrane-specific measurements. There are a few online readings of local flue-gas composition, membrane-side pressure drops, or signs of fouling. Recent reviews on DT standardization have highlighted that the lack of common data models, interfaces, and terms for sharing high-frequency membrane and plant data is a significant obstacle to creating interoperable DSs and DTs [48,102]. Third, membrane-based CO_2 capture remains in the early stages of scale-up. Many systems operate at the pilot or demonstration level, resulting in short, fragmented datasets. The high computational cost of CFD-resolved multiphase models further limits the possibility of continuously synchronized, full-fidelity DSs and DTs for CO_2 separation modules. A practical way forward is to stage the digitalization process rather than aiming immediately for a complete plant-wide DT. A useful first step is to create hybrid surrogates and DS layers around existing process-simulation environments. These would use reduced-order CFD or ML surrogates trained on high-fidelity simulations and design studies [103]. In the next step, the same systems that have proven effective in processes such as UF or RO water treatment processes, and MBR DTs, where DTs help with energy-optimized control, predictive fouling management, and remote monitoring, can be applied to membrane-based CO_2 capture with few structural changes. Only the process and property models will differ [43,77,93]. Finally, as more capture plants start using standard industrial communication protocols, such as open platform communications unified architecture (OPC UA)-based historians and edge gateways, and as AI-driven CCS platforms for solvent and adsorption systems improve, membrane-specific DS and DT modules can be integrated into plant-wide CCS for better coordination of capture efficiency, compression, and storage operations [104]. Although DS and DT frameworks for membrane CO_2 capture have not yet been demonstrated, enabling components, as shown in Fig. 15, exist.

These components form the building blocks of future DS- and DT-enabled membrane CO_2 capture systems. Hybrid CFD–ML surrogates and AI-assisted membrane design have shown <10% prediction error in transport properties and up to ~20% improvement in energy-efficiency in process simulations. Recent work demonstrated generative AI for ionic-liquid membranes and ML-screening of MOF-based MMMs, accelerating material discovery and enabling real-time optimization concepts [105–108].

Table 3 presents the researches exploring hybrid modelings applied to membrane systems for CO_2 capture is.

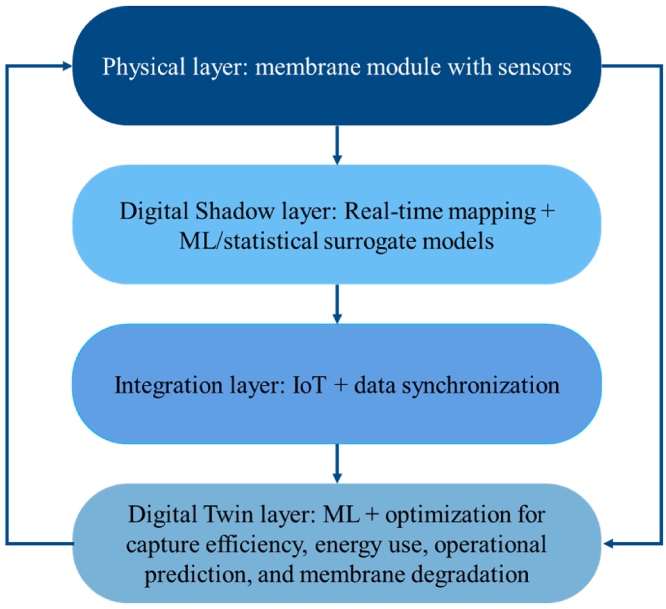


Fig. 15. Components of membrane-based CO_2 capture for the digitalization process.

3.3.2. *Hydrogen purification process*

To address the challenges posed by climate change, identifying renewable and clean energy alternatives is of great importance. Numerous studies have demonstrated that H_2 is a promising solution [112–115]. H_2 can be produced via various routes, particularly via methane steam reforming. Nevertheless, there are challenges regarding the safe production and storage of this substance. In addition, some of the main processes are restricted by equilibrium limitations [116–118]. Membrane-based processes can overcome these challenges by shifting reaction equilibria toward higher H_2 yields [119–122]. Investigating these systems for H_2 purification is crucial to achieving high efficiency while maintaining safety. Several studies have examined these systems using mathematical and computational approaches. However, such methods can be time-consuming and costly. With the advancement of AI-driven and DT frameworks, these approaches have gained considerable attention in H_2 separation technologies [123–126].

Recent studies have leveraged hybrid models and DT techniques to predict, optimize, and control H_2 separation in membrane-based systems. Table 4 summarizes recent works in this area.

At the same time, Fig. 16 illustrates the overall schematic of the hybrid modeling framework proposed by Torabi et al. [129,130], which integrates prediction and optimization for H_2 production and purification in packed bed membrane reactors (PBMRs).

Despite these recent advancements, a notable gap remains in the application of DS to membrane-based H_2 separation. Consequently, current research efforts rely predominantly on offline ML models or fully simulated DT frameworks, leaving DS largely unexplored. Closing this gap would provide a valuable intermediate layer toward full digitalization of H_2 purification systems, enabling real-time process insights, early fault detection, and adaptive operational recommendations at substantially lower computational cost than DTs.

Overall, Table 5 shows the potential of DTs and hybrid models to significantly enhance H_2 production through improved process control, predictive maintenance, and cost optimization. While most reported studies remain at the laboratory or pilot scale, they provide convincing evidence that DT frameworks can bridge the gap between physics-based modeling and real-time industrial deployment. Nevertheless, further research is required to address challenges such as data scarcity, model generalization across scales, and integration with plant-wide energy systems.

Table 3
Hybrid modeling studies on design, operation, and optimization of CO_2 capture using membrane-based systems.

Process	Modeling Approach	ML Architecture	Hybridization	Generalization/Physical Consistency Approach	Ref.
MOF and MMMs	Molecular simulation + ML-based property prediction	Multiple regression pipelines including LASSO-LARS cross-validation (CV) (for He adsorption), Extra Trees regressor (ETR) (for H_2/CH_4), GB regressor (GBR) (for N_2 adsorption/diffusion), RF/GBR/ETR variants with polynomial feature expansion	Molecular simulations computed adsorption/diffusion of gases in >100k MOF structures; ML models mapped structure descriptors (LCD, PLD, density, metal%, O–to–M, TDU, DU) to transport properties. ML replaced repeated molecular simulation when exploring MMM design.	Strong multicollinearity in geometric/chemical descriptors was removed using Pearson correlation filtering (>0.9 removed). Five-fold cross-validation controlled overfitting. ETR and GBR reduced computational cost while maintaining physical consistency by learning monotonic structure–performance trends verified in simulation plots.	[109]
Key Technical Insights (Advantages/Limitations): Ensemble models (ETR, GBR) achieved the highest accuracy in predicting permeability and diffusivity by capturing nonlinear structure–transport relationships. However, models trained solely on simulation data lacked a guarantee of physical behavior outside the MOF domain. Generalization depended strongly on the diversity of simulation-derived feature distributions.					
Ionic-liquid (IL)-incorporated MOF fillers in MMMs	High-throughput molecular simulation + ML screening	RF with feature elimination, Pearson correlation filtering, five-fold CV; trained on 3593 IL-MOF composites	GCMC + MD simulations produced CO_2/N_2 adsorption and diffusivity; RF predicted CO_2 permeability and CO_2/N_2 selectivity using structural descriptors (PLD, vASA, gASA, AV, density, Vfree + chemical descriptors). RF extracted structure–performance rules (e.g., AV → selectivity; gASA → permeability).	Feature multicollinearity was removed (threshold = 0.9). Model used 90/10 train–test split and five-fold CV. RF was computationally cheap compared to full GCMC/MD for 8000+ composites. Physical consistency was indirectly ensured through descriptor filtering (only MOFs satisfying pore-size constraints > N_2 kinetic diameter were kept).	[106]
Key Technical Insights (Advantages/Limitations): RF accurately identified IL-MOF fillers with optimal CO_2 transport. Advantage: clear feature importance allowed mechanistic understanding of why IL improved affinity/transport. Limitation: RF did not encode thermodynamic laws explicitly; extrapolation beyond CoRE MOF distribution may fail.					
Large-scale screening of experimental MOF membranes	GCMC/MD simulations + ML-based structure descriptor ranking	Seven ML classifiers tested; eXtreme Gradient Boosting (XGBoost) selected as best performer for descriptor importance ranking	Molecular simulations produced $CO_2/N_2/O_2$ permeability. ML classified MOFs into high- and low-performance classes. XGBoost ranked six key descriptors (LCD, PLD, VSA, φ, density, PSD% (2.5–3.5)) controlling permeance. Provided mechanistic interpretation: available permeation area dominated by small-kinetic-diameter gases.	A train/test split was used; XGBoost mitigated overfitting through tree regularization and shrinkage. The computational cost was low relative to GCMC/MD for thousands of structures. Physical consistency is maintained because XGBoost feature contributions are aligned with univariate structure–permeance trends.	[110]
Key Technical Insights (Advantages/Limitations): XGBoost successfully identified top-performing MOFs and revealed physically interpretable transport-controlling features. Limitation: ML only classified membranes; it did not generate continuous surrogate models or uncertainty estimates, limiting use for process integration.					
Multi-stage CO_2 capture process with membrane units	Process simulation + surrogate ML to accelerate design	Gaussian surrogate regression and polynomial response surfaces replacing detailed adsorption/membrane submodels.	Aspen simulations provided high-fidelity reactor/membrane data; surrogates mapped (T, P, feed ratios) → CO_2 capture performance. Surrogates replaced computationally expensive equilibrium/kinetic modules in multi-stage design optimization.	Surrogates reduced simulation time drastically; CV checked interpolation accuracy. However, surrogate fidelity depended on the density of sampling points. Physical consistency was enforced by generating all training data from mechanistic simulations.	[111]
Key Technical Insights (Advantages/Limitations): Advantage: massive acceleration in multi-unit CO_2 capture process design, enabling large search spaces. Limitation: surrogate broke down when extrapolating beyond sampled regions; no embedded mass/energy balance constraints inside ML model.					

The future of AI-driven H_2 production is expected to advance rapidly through the adoption of AI-enabled DSs and DTs, which provide virtual replicas of H_2 plants to support real-time monitoring, predictive maintenance, and process optimization [138].

Beyond CO_2 and H_2, hybrid models, DS-, and DT-enabled membrane design has been applied to air separation and acid-gas removal. Computer-aided polymer design models have successfully identified new materials for O_2/N_2 membranes, showing improved permeability-selectivity trends [139,140]. Studies regarding the application of hybrid models and DTs for membrane-based gas separation processes are presented in Table 5. However, research in these areas remains limited and requires further attention.

Overall, the case studies on gas separation membranes consistently employ ML-assisted or hybrid modeling frameworks to improve permeability–selectivity trade-offs. While most studies focus on performance enhancement and process optimization, limitations related to data availability and scale-up remain common across different gas separation applications.

3.4. Catalytic membrane reactors

A catalytic membrane reactor (CMR) integrates chemical reactions with membrane-based separation in a single intensified unit, where reaction and separation occur simultaneously and synergistically [141]. In these systems, the membrane functions dually, serving as both a selective barrier and an active component that influences reaction kinetics. This innovative concept has attracted considerable interest in diverse fields such as chemical and petrochemical industries, biotechnology, and environmental engineering, owing to its exceptional ability to intensify processes, enhance efficiency, lower energy demands, and reduce operating costs compared to traditional reactor configurations [9,11,142–145].

Compared to water treatment and gas separation membranes, the application of digital twins and hybrid models in catalytic membrane reactors remains at an early stage. Hybrid models are also emerging in CMRs, enabling practical predictions of simultaneous reaction–separation processes. Studies on Pd- and silica-based reactors demonstrate

Table 4
Studies on DT and hybrid modeling for the design, operation, and optimization of H_2 purification using membrane-based systems.

System/Membrane Type	Goal	Gas/Feed	Approach	What the ML/modeling component does	Ref.
Chemical looping H_2 production system	Real-time control/ Optimization	O_2 carrier–gas systems	DT	ML-driven DT aligned gas flow rates with O_2-carrier reactivity in real-time; the GB decision tree model achieved an R^2 of 0.87 for H_2 yield prediction and adaptive flow control within 10 s.	[127]
Renewable-powered membrane-based electrolysis system	Process Optimization/ Adaptive control	$H_2O \rightarrow H_2 + O_2$	DT	Used DT for self-optimizing control, predictive maintenance, and resilient renewable-grid coupling in large-scale electrolysis.	[128]
Pd–Ag packed-bed membrane reactor (PBMR) for biogas tri-reforming → H_2	Design/Operation/ Optimization	$CH_4/CO_2/H_2O/O_2 \rightarrow H_2$	Hybrid Modeling	The ML surrogate, trained on CFD, enabled rapid exploration of the operating/design space.	[129]
Methylcyclohexane (MCH) (LOHC) dehydrogenation PBMR → H_2	Design/Operation/ Optimization	MCH → H_2 + toluene	Hybrid Modeling	ML surrogate-derived set-point selection/optimization around a mechanistic model	[130]
Proton exchange membrane water electrolysis	Parameter estimation/ Thermal management	$H_2O \rightarrow H_2 + O_2$	DT	Integrated electrochemical and heat-transfer models for sensor-less temperature monitoring and improved stability.	[131]
High-temperature proton exchange membrane electrolyzer cell	Dynamic behavior prediction/Control	$H_2O \rightarrow H_2 + O_2$	DT	Combined Multiphysics simulation and nonlinear system identification; neural network control reduces power overshoot and improves durability.	[132]
Green H_2 production facilities (techno-economic DT)	Design/Investment optimization	–	DT	Monte Carlo–based DT assessed cost, reliability, and risk, optimizing design for the internal rate of return and investment trade-offs under uncertainty.	[133]
Materials: Palladium-alloy H_2 membranes (screening)	Design/Optimization	H_2/impurity mixtures	Hybrid Modeling	First-principles calculations integrated with ML accelerated prediction of key separation properties of Pd-alloy membranes.	[134]

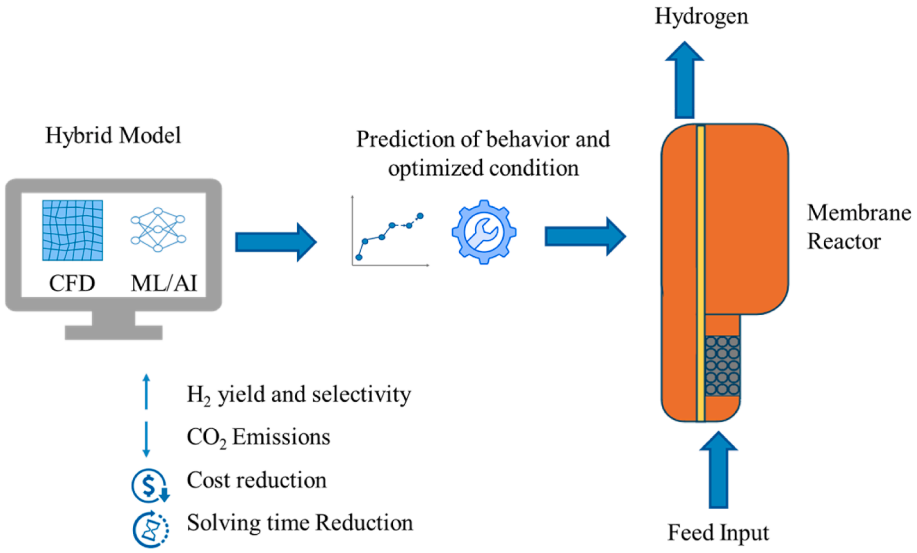


Fig. 16. Hybrid model framework used in Torabi et al.'s work [129,130].

improved H_2 yield prediction accuracy (<5–8% error) and >70% reduction in simulation time by coupling mechanistic models with ANN-based surrogates [146].

Despite these advances, a clear gap remains in the adoption of DS and DT frameworks for CMRs. While hybrid modeling studies are emerging, no existing research has implemented DS and DT for Pd-, silica-, or ceramic-based CMR systems.

This gap can be due to the complex nature of these systems, since CMRs couple heterogeneous reaction kinetics, multicomponent diffusion, permeation, and catalyst deactivation in highly intensified, often non-ideal geometries. Constructing a DS or DT that resolves these multiphysics interactions in real time, while accounting for conservation laws and membrane selectivity, requires detailed models that are computationally intensive and difficult to calibrate under plant disturbances [48,147]. In addition, most reported CMR studies remain at laboratory or pilot scale, where instrumentation typically limits measurements to inlet–outlet flows and compositions; internal states (e.g., local reaction rates, shell-side concentration polarization, or membrane aging) are not directly measured, which restricts observability and precludes robust online state estimation. From a systems perspective, CMR prototypes are often operated as stand-alone test rigs without fully developed distributed control systems and supervisory control and data acquisition (SCADA) integration, making it difficult to implement the continuous data streaming and standardized data interfaces that DT platforms require [148–150].

A realistic near-term strategy to address this gap is to leverage mature DS and DT approaches from related catalytic and bioprocess systems. Recent work on pilot-scale catalytic reforming, fermenters, and generic chemical reactors shows that modular mechanistic core models combined with reduced-order or ML-based surrogates can be embedded into flexible DS and DT architectures for different reactor designs [151–153]. Transferring these concepts to CMRs suggests a staged roadmap: (i) construct hybrid models in which detailed 1D/2D mechanistic CMR models are used offline to generate synthetic datasets, on top of which fast surrogates (e.g., gradient-boosted trees, shallow ANNs, or physics-informed ROMs) approximate conversion, selectivity, H_2 recovery, and temperature profiles; (ii) wrap these surrogates into a DS that passively receives real-time measurements from CMR test rigs for soft sensing (virtual sensors of internal states), anomaly detection, and what-if analysis; and (iii) upgrade selected DS implementations into full DTs by closing the feedback loop,

Table 5
AI-driven studies on the design, operation, and optimization of gas separation using membrane-based systems.

Process	Modeling Approach	ML Architecture	Hybridization	Generalization/Physical Consistency Approach	Ref.
Polymeric membranes	Hybrid ML + Computer-aided molecular design (CAMD) for polymer discovery	GBR, RF, LASSO, and feature-importance methods embedded in the CAMD framework	ML models learned structure–property relationships (permeability, diffusivity) from polymer datasets; CAMD then used these ML-derived rules as constraints inside group-contribution physics-based design equations. ML results restricted CAMD search space to chemically feasible polymer designs with target O_2/N_2 performance.	Multicollinearity was removed using VIF and Pearson filtering. Five-fold CV prevented overfitting. Physical consistency was enforced by requiring ML-suggested polymer fragments to satisfy CAMD's mechanistic constraints (e.g., solubility parameters, functional-group additivity). Computational cost decreased because ML replaced thousands of expensive molecular-simulation evaluations.	[135]
Key Technical Insights (Advantages/Limitations): Advantage: ML generated interpretable structure–performance links that improved CAMD feasibility filters; hybridization maintained physical realism. Limitation: model accuracy depended strongly on training-set coverage; extrapolation to unconventional polymer chemistries remained uncertain.					
Spiral-wound gas membrane module	DT-oriented monitoring framework using data reconciliation (soft sensor forming basis for DT)	No predictive ML model; instead, a statistical soft-sensor model (data reconciliation + gross-error detection) was used for real-time estimation and anomaly detection	Phenomenological membrane model and mass-balance constraints reconciled plant measurements; the soft sensor estimated unmeasured variables and corrected biased signals. This reconciliation procedure formed the "physics layer" necessary for a future DT but did not include a machine-learning regression component.	Weighted least squares (WLS) ensured consistency with conservation laws. Gross-error detection prevented divergence. Computational cost was minimal, allowing real-time use in industrial plants. Physical consistency was guaranteed because all reconciled variables satisfied mass-balance equations.	[136]
Key Technical Insights (Advantages/Limitations): Advantage: provided reliable real-time monitoring and estimation of unmeasured variables; directly enabled DT deployment. Limitation: lacked predictive ML algorithms and could not evaluate nonlinear membrane behavior beyond the mechanistic model.					
Tubular gas-separation membranes	Hybrid CFD + ML surrogate; used as core of DT-ready AI framework	XGBoost (tree-boosting; hyperparameters tuned via 10-fold CV and grid search)	A validated 3D CFD model generated 1000+ simulations via Latin Hypercube Sampling (LHS). XGBoost learned the mapping: operating conditions + geometry + membrane properties → recovery, purity, cost. The surrogate replaced expensive CFD inside optimization and DT scenarios.	XGBoost prevented overfitting via regularization (λ), shrinkage (learning_rate), and subsampling; 10-fold CV assured generalization. Computational cost dropped from hours per CFD simulation to milliseconds per surrogate prediction. Physical consistency was indirectly preserved because the surrogate's training set consisted entirely of CFD solutions obeying conservation and transport laws.	[96]
Key Technical Insights (Advantages/Limitations): Advantage: extremely fast predictions, enabling >100,000 designs in minutes; high accuracy (RMSE ≈ 3–5%). Limitation: surrogate failed when extrapolated beyond CFD sampling ranges; did not embed physical constraints explicitly (only implicitly through training data).					
Porous media/ membrane contactor for gas–liquid or gas–gas separation	Hybrid CFD + ML ensemble modeling of spatial concentration fields	DT, ET, RF, Histogram-based Gradient Boosting (HBGB). ET achieved best accuracy ($R^2 \approx 0.9967$).	A CFD mass-transfer model generated >25,000 spatial samples (r,z → C). ML predicted concentration fields from spatial coordinates, replacing CFD when evaluating transport phenomena across design variations.	Outlier removal via Cook's distance improved stability. Successive Halving optimized hyperparameters efficiently. Ensemble methods handled nonlinear CFD outputs with low variance. Physical consistency was indirectly maintained since training data obeyed diffusion-convection equations.	[137]
Key Technical Insights (Advantages/Limitations): Advantage: ET and RF captured spatial gradients accurately with very low error; prediction cost was negligible compared to CFD. Limitation: ML models had no embedded physics and could misrepresent behavior outside the CFD domain; results sensitive to CFD mesh quality.					

linking the virtual CMR to supervisory control layers (e.g., model-predictive control) for adaptive optimization of feed composition, sweep-gas flow, and operating temperature under constraints on membrane integrity and catalyst life. As standardized DT toolchains and reactor-agnostic DT frameworks for the chemical industry continue to evolve [48,149,154], they can be directly repurposed for CMRs by replacing the unit-operation library with membrane-specific submodels and by defining consistent data schemas for permeate and retentate variables and degradation indicators. Table 6 summarizes recent studies employing hybrid modeling for the design, monitoring, control, and optimization of CMR systems. This contrast highlights a clear opportunity for future comparative and cross-domain digitalization strategies.

Table 6
Studies using AI for the design, operation, and optimization of catalytic MR systems.

System/Membrane Type	Goal	Gas/Feed	Hybrid/DT approach	What the ML/modeling component does	Ref.
WGS CMR (Pd-based)	Design/Operation/ Optimization	$CO + H_2O \rightarrow CO_2 + H_2$ (WGS)	Hybrid Modeling	ANN trained on mechanistic MR data to predict performance and guide operating conditions	[155]
Silica CMR for methanol steam reforming	Design	$CH_3OH + H_2O \rightarrow CO_2 + H_2$	Hybrid Modeling	Mechanistic model integrated with ANN modeled MR performance (conversion, H_2 yield) vs. key variables	[156]
Pd–Ag CMR for glycerol steam reforming	Design/Operation/ Optimization	Glycerol + $H_2O \rightarrow H_2 + \ldots$	Hybrid Modeling	Mechanistic model coupled with ANN predicted conversions/selectivity in Pd–Ag CMR; identified influential factors	[157]
Membrane-assisted reactor for bio-alcohols	Design	Methanol/ethanol reforming routes → H_2	Hybrid Modeling	Hybrid model (with MR stage) complemented the math model to enhance H_2 prediction/analysis	[158]
MR for methanol steam reforming (Fuel-cell grade H_2)	Design/Screen	$CH_3OH + H_2O \rightarrow CO_2 + H_2$	Hybrid Modeling	ANN predicted system behavior for MR; accelerates screening of conditions	[159]

3.5. Fuel cells

Proton exchange membrane fuel cells (PEMFCs) have emerged as one of the most promising options for clean energy conversion due to their high efficiency, rapid response, and suitability for portable, transportation, and stationary power applications. At the core of a PEMFC is a dense proton-conducting polymer membrane that enables proton transport from the anode to the cathode. H_2 is catalytically oxidized at the anode to produce protons, which migrate through the membrane while electrons flow through an external circuit; at the cathode, these protons react with O_2 to generate water and heat [160].

Despite their advantages, PEMFCs still require significant improvements in cost, performance, and long-term durability to meet large-scale commercialization targets [161] Given the complex and tightly coupled physical and chemical processes occurring within the cell, the development of AI-driven models, DSs, and DTs presents an important opportunity to enhance system design, operational monitoring, and intelligent control [24,162]. Hybrid models combining mechanistic and data-driven elements can support the prediction of critical factors governing fuel cell performance—including durability, electrocatalyst degradation, water and thermal management, dynamic load response, and cold-start behavior [162]. DSs and DTs further strengthen this capability by enabling high-fidelity virtual representations of fuel cells that simulate internal multi-physics processes, generate diverse fault scenarios, and predict state evolution in real time. When integrated with data-driven learning, DTs enable more accurate fault detection, improved diagnostic robustness under limited data, and enhanced adaptability across devices and operating conditions. Consequently, DT-enabled diagnosis offers a promising pathway toward more resilient, efficient, and commercially viable PEMFC systems [163,164]. Published studies show that DT–based diagnostic frameworks can achieve exceptionally high accuracy for PEMFC fault detection, with reported performance exceeding 99% for faults such as pressure loss, membrane drying, reactant flow disruption, and flooding [163]. Similarly, hybrid modeling approaches that have demonstrated strong predictive capability, achieving coefficients of determination (R^2) above 0.99 for key PEMFC attributes and internal states under a wide range of operating conditions [165]. Table 7 presents the studies that have applied these frameworks to PEMFC systems. Overall, the findings highlight the strong potential of DSs, DTs, and hybrid models to enable highly accurate fault diagnosis and predictive monitoring in PEMFC applications.

4. Role of computational fluid dynamics (CFD) in hybrid ML-based development

CFD, as a high-fidelity mechanistic modeling tool, provides a valuable foundation for developing hybrid models and can also be incorporated into DS and DT frameworks when detailed physical insight is required. Although CFD is not essential for all digital solutions, it adds real value by creating virtual models of membrane processes that resolve mass, momentum, and energy transport phenomena with high spatial and temporal accuracy. Explicitly capturing real module geometries, flow structures, and fluid–membrane interactions helps to clarify hydrodynamics, operating conditions, and membrane configuration effects without interfering with the actual process stream. These simulations offer mechanistic insight into separation performance, concentration polarization, and fouling development, serving as a physics-based complement to data-driven models within hybrid, DS, and DT architectures [175].

While CFD is widely used to analyze mass transfer and coupled transport phenomena in membrane modules, its computational demands limit its use in applications requiring rapid response or continuous updating. This limitation has accelerated interest in data-driven approaches, such as ML, which offer more rapid prediction and easier integration into online environments. By training ML models on data generated from CFD, experiments, or operational measurements, hybrid frameworks can reproduce key transport and fouling behaviors at a fraction of the computational cost, thereby enabling efficient and reliable evaluation of membrane processes (see Fig. 17) [176].

These hybrid models combine physics-based structure with learned behavior, creating fast surrogate models suitable for integration into real-time DS and DT systems. Their ability to approximate nonlinear transport phenomena, fouling dynamics, and reaction–separation interactions make them important building blocks for digital membrane technologies [44,166–169]. When paired with real-time sensor data and AI-based updating mechanisms, CFD-derived surrogates can be embedded within DS and DT architectures (see Fig. 18), enhancing predictive accuracy, and enabling dynamic adaptation to operational variations. Purely data-driven models often require large training datasets and may suffer from overfitting, limited extrapolation capability, and lack of interpretability.

In contrast, mechanistic models such as CFD provide structured physical constraints and more robust extrapolation, provided that their assumptions remain valid. Recognizing the distinct strengths of these approaches has led to the adoption of hybrid frameworks in DS and DT systems, combining mechanistic fidelity and data-driven adaptability to produce semi-parametric models with improved accuracy and reduced data requirements [177–180]. Unlike conventional simulations, DSs and DTs represent the actual operating state and can test optimal operating strategies or maintenance actions in silico before implementing them in the plant. For example, CFD-based simulations of concentration polarization in filtration modules have been used to generate training data for AI models that dynamically adjust flow conditions to minimize fouling and extend membrane lifetime [43,67]. Such integration demonstrates how CFD can act as a physics engine within DS and DT frameworks, ensuring mechanistic reliability while leveraging data-driven adaptability for real-time process control. Table 8 presents the advantages and disadvantages of the models mentioned.

Table 7
Summary of previous studies on hybrid modeling, DS, and DT for PEMFCs.

Category/System	ML Architecture	How Physical + Data-Driven Modeling Were Combined	Generalization/Physical Consistency Approach	Key Technical Insights (Advantages/Limitations)	Ref.
Hybrid modeling/140-cell PEMFC stack voltage and uniformity prediction	Ensemble ML models: RF, GB, SVM, and shallow ANN architectures (trained with dynamic cycling data)	ML models predicted cell-wise voltage from physical operating conditions (I, T, humidity, stoichiometry); stack-level electrochemical relationships (Nernst & Ohmic laws) constrained feature space.	K-fold CV reduced overfitting; temporal smoothing enforced physically plausible voltage transitions; computational cost remained moderate as models were shallow.	Advantage: robust voltage prediction (<18 mV error). Limitation: models lacked explicit degradation physics → risked drift during long-term cycling.	[166]
Hybrid modeling/Pt-loading effects on PEMFC voltage	ANN (single and double hidden-layer networks) + empirical electrochemical model (activation and ohmic losses)	ANN estimated complex Pt-loading-dependent activation losses; physical stack model computed Nernst, ohmic, and mass-transport terms. ANN provided residual correction on top of the physics model.	Weight decay + early stopping prevented ANN overfitting; physical constraints ensured monotonic voltage-current behavior. Computational cost remained low.	Advantage: ANN-physics hybrid significantly outperformed pure physics; limitation: ANN accuracy deteriorated under unseen humidity–temperature combinations.	[167]
DS/Internal PEMFC field reconstruction (T, humidity, flooding)	Deep CNN encoder-decoder (U-Net style) trained on multiphysics datasets	CNNs were trained to map external, measurable variables (voltage, pressure, humidity) to internal fields generated by a 3D multiphysics model. Served as a DS of internal states.	Data augmentation improved generalization; spatial convolutions preserved physics-consistent spatial correlations; inference cost was extremely low.	Advantage: high-quality field reconstruction (similarity ≈0.96). Limitation: DS performance degraded when operating conditions deviated from those in the training scenarios.	[168]
DS/RUL prediction using stacked autoencoders (SAE) + transfer learning	SAE + domain-adaptation transfer learning + deep MLP regressor	SAE extracted degradation features; transfer learning adapted DS to new operating regimes with limited data; DS updated RUL estimates based on online voltage decay.	Transfer learning reduced reliance on large datasets; SAE compression reduced computational cost; and physical consistency was maintained under a monotonic-degradation assumption.	Advantage: high RUL accuracy under limited sensing. Limitation: SAE latent features lacked direct interpretability of electrochemical degradation processes.	[169]
DT/Cross-device PEMFC fault diagnosis	Transfer-learning CNN + domain-adaptation network + DT-generated synthetic faults	Multi-physics DT generated fault signatures; ML classifier trained on these, then fine-tuned with small real datasets. Physics ensured realistic fault dynamics (drying/flooding/pressure loss).	Domain adaptation ensured generalization across devices; the synthetic-to-real shift reduced data requirements; computational cost was moderate.	Advantage: ≈99% fault accuracy. Limitation: Synthetic data is limited by the fidelity of the DT physical model.	[163]
DT/Multi-physics-resolved PEMFC DT (3D physical model + surrogate)	Physics-based model + DNN surrogate model (feed-forward regression)	High-fidelity CFD/electrochemical model produced training data; DNN surrogate predicted internal fields (T, λ, O_2/H_2 distribution).	Surrogate reduced computation time by orders of magnitude; physical consistency ensured by training exclusively on 3D model solutions.	Advantage: substantial speed-up for internal-field prediction. Limitation: Surrogate lost accuracy (4–25% error) in high-nonlinearity zones.	[24]
DT/PEMFC fault diagnosis without fault labels	Unsupervised domain-adaptive learning (adversarial neural network) + DT-generated faults	The DT model produced synthetic faults; unlabeled real data were used to align distributions via domain adaptation; enabled label-free fault detection.	Reduced labeling cost; adversarial training enhanced generalization across domains; moderate computational cost.	Advantage: 92–93% accuracy with zero real fault labels. Limitation: performance depended on DT–real domain closeness.	[170]
DT/Dynamic load optimization & thermal management for automotive PEMFC	ML-assisted multi-objective optimization using NSGA-II + data-driven stack model (DNN surrogate)	DT included a high-fidelity Simulink model; a surrogate predicted the stack response at each optimization step; and NSGA-II optimized stoichiometry, hydration, and cooling rate.	Surrogate reduced computational cost by ~29%; multi-objective optimization balanced efficiency vs. durability; CV mitigated overfitting.	Advantage: ~20% improved transient response; ~38% reduction in model-experiment error. Limitation: Surrogate accuracy is limited outside the explored parameter domain.	[171]
DT/MPPT control using GPR + DT	GPR + Sliding-Mode Controller (SMC) trained using DT-generated data	DT of PEMFC + converter generated a large dataset; GPR predicted optimal duty cycle near MPP; SMC refined final control.	GPR provided uncertainty-aware predictions; training cost was moderate; DT ensured physics-consistent MPP dataset.	Advantage: 67% improvement at start-up; 43% improvement in load change. Limitation: GPR computation is computationally more expensive than ANN for large datasets.	[172]
DT/Self-adaptive DT for RUL prediction	Deep CNN + quantile Huber-loss regression + transfer learning	CNN extracted health indicators; DT updated the degradation trajectory via domain adaptation; physical monotonicity was enforced via a custom loss.	Quantile Huber-loss improved robustness; transfer learning improved generalization across operating profiles; cost was moderate.	Advantage: 0.94–0.96 accuracy across devices. Limitation: CNN latent features remain non-physically interpretable.	[173]
DT/PEMFC aging estimation & synthetic aging-data generation	Mechanistic degradation model + calibrated parameter-estimation ML layer (regression + nonlinear curve fitting)	DT integrated degradation physics (Pt dissolution, carbon corrosion). ML tuned parameters based on experiments; DT generated synthetic long-term datasets.	Physics ensured accurate degradation trends; ML accelerated calibration; and data generation reduced experimental costs.	Advantage: <5% deviation vs. experiments. Limitation: accuracy is limited by the correctness of embedded degradation mechanisms.	[174]
DT High-temperature PEM electrolyzer (HT-PEMEC) DT	Hammerstein-Wiener nonlinear system identification + fuzzy control + neural network predictive control (NNPC)	2D multi-physics model generated a dynamic dataset; nonlinear system ID built a surrogate; DT used for control strategy design (fuzzy + NNPC).	Surrogate reduced simulation cost; physical consistency guaranteed by using multi-physics training data; generalization depended on input nonlinearities.	Advantage: NNPC reduced power overshoot by ~92%. Limitation: Surrogate accuracy decreased for large thermal transients.	[132]

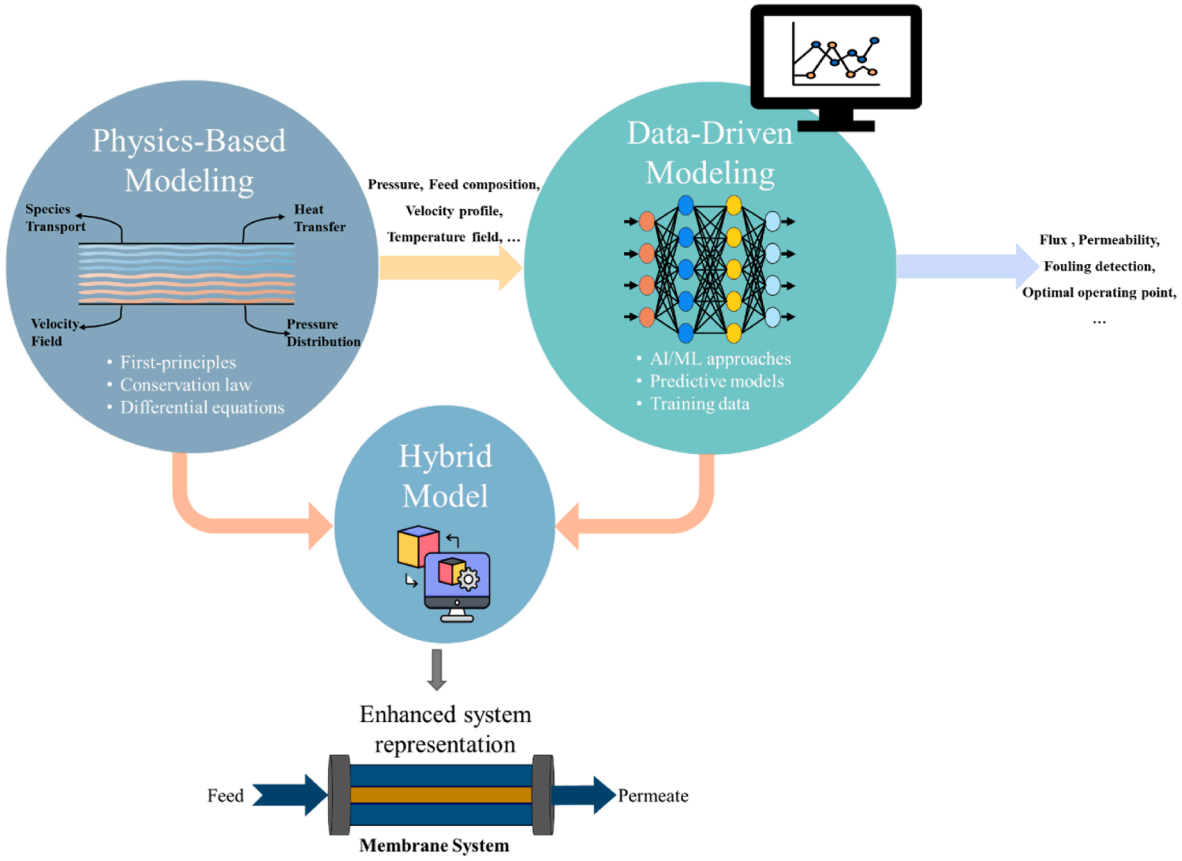


Fig. 17. Combining physics-based and data-driven modeling approaches into a hybrid model for membrane systems.

5. Challenges and limitations

Implementing DS and DT in membrane processes offers powerful opportunities for real-time monitoring, soft sensing, predictive control, and optimization. However, despite their strong potential, adoption in membrane-based systems remains limited. Several persistent challenges must be addressed before DS and DT frameworks can become technically robust, economically feasible, and widely deployable.

A key barrier is the lack of clear concepts and standardization in the membrane community. To address this, the review provides standardized definitions that clearly distinguish unidirectional DS from bidirectional DT. Additionally, validated tools or guidelines specific to this area are absent. While industries such as manufacturing and aerospace have begun adopting unified DT quality-assurance frameworks, no equivalent protocols exist for membrane applications. This lack of clarity limits reproducibility, interoperability, and the ability to integrate DS and DT modules with current membrane monitoring and control systems [25,48,181]. As noted by Milisavljevic-Syed et al. [148], DT validation currently lacks harmonized metrics and cross-platform test procedures, limiting reproducibility and comparability across studies. Similarly, Pal et al. [149] emphasized that process-industry DTs require standardized model-validation pipelines, data schemas, and performance indicators before widespread deployment can be achieved. At present, membrane DS and DT studies validate their models using case-specific criteria, such as prediction error, flux deviation, or fouling-detection accuracy, without a unified methodology for assessing digital fidelity, robustness under disturbances, sensor-drift resistance, or scalability across modules. This limitation prevents meaningful benchmarking across different membrane configurations (e.g., RO, MBR, and gas-separation modules) and hinders technology transfer from pilot studies to industrial-scale systems. Establishing domain-specific DS and DT validation standards that incorporate multi-level metrics (module-scale, system-scale, and lifecycle-scale), uncertainty quantification protocols, and stress-testing procedures under realistic operating conditions is therefore an essential research priority for the field.

Another major limitation is the technical scalability of DS and DT deployment in membrane systems. As membrane plants grow, whether through additional pressure vessels, multi-stage RO trains, large MBR basins, or cascaded gas-separation modules, the digital model must encompass an increasing number of interacting subsystems. This dramatically increases computational load, model-update frequency, and the volume of incoming sensor data. In DS frameworks, scalability affects the ability to maintain consistent real-time tracking across multiple distributed units. In contrast, DTs challenge the feasibility of synchronizing multiple high-fidelity models with live operational data. Moreover, membrane aging, fouling progression, and material degradation introduce dynamic changes (i.e., model drift) that require continuous recalibration or adaptive updating, which becomes increasingly complex as system size and complexity grow [48,169,182].

Data-related constraints further hinder implementation. Both DS and DT depend on continuous, high-quality sensor streams; however, membrane systems frequently operate with sparse, noisy, low-frequency, or low-resolution measurements (e.g., inlet/outlet pressure, temperature, bulk permeate flow), while critical internal states such as local fouling resistance, concentration polarization, membrane compaction, and spatial flux maldistribution are typically unmeasured. This creates low-observability conditions that hinder accurate state estimation and real-time model updating [183]. Additionally, sensor drift, calibration errors, scaling effects, and fouling-induced signal fluctuations introduce non-stationary noise that degrades prediction accuracy and increases model drift frequency in DS and DT deployments [184–187]. In many facilities, particularly older RO, UF, and MBR plants, the instrumentation layer is based on legacy PLC/SCADA architectures with limited support for synchronized high-bandwidth

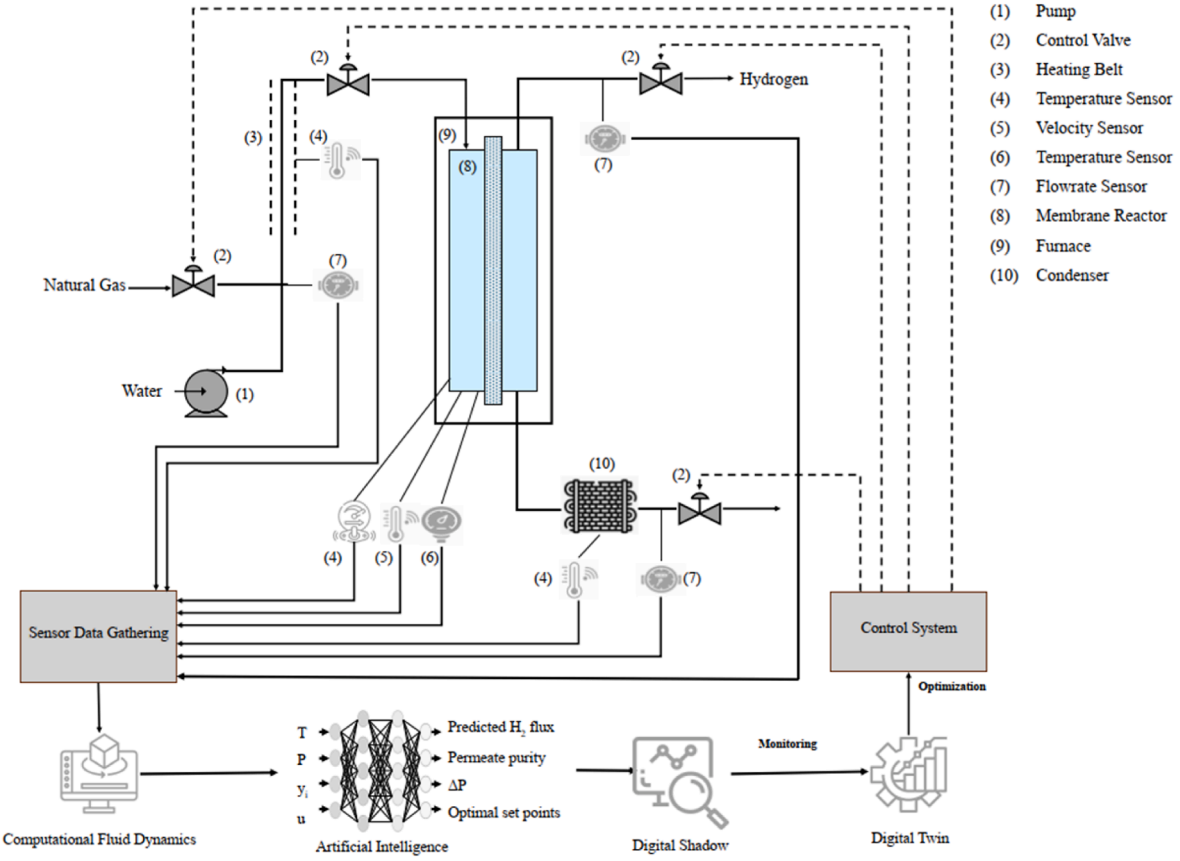


Fig. 18. Hybrid integration of CFD and AI models for DS and DT development in membrane processes, allowing real-time monitoring and optimized performance.

streaming, cybersecure data pipelines, or standardized data formats (e.g., OPC UA), all of which are prerequisites for scalable DT systems [188]. The situation is even more restrictive for gas-separation units and CMRs, where high-temperature, corrosive, or multiphase environments make long-term sensor survival difficult, resulting in frequent data gaps and unreliable readings [189,190]. To address these barriers, several studies recommend a data-centric digitalization pathway: (1) deployment of robust sensor suites (fiber-optic pressure sensors, multi-parameter probes, corrosion-resistant gas analyzers, low-fouling optical sensors); (2) use of redundant sensing and in-situ calibration to mitigate drift and noise; (3) integration of edge computing and industrial IoT gateways to preprocess data, manage uncertainty, and compress high-frequency signals before transmission; and (4) incorporation of hybrid observers and virtual sensors within DS frameworks to infer unmeasured membrane states (e.g., fouling resistance, mass-transfer coefficients, local polarization indices) [43].

Together, upgrading the sensing infrastructure and implementing data-quality governance provides a realistic pathway to enable reliable, stable, and maintainable DS and DT development in membrane separation plants.

Integration with existing membrane infrastructure presents additional challenges. Many membrane facilities, especially municipal wastewater treatment plants, seawater desalination plants, and industrial gas separation units, rely on legacy equipment that lacks built-in IoT capability or instrumentation ports. Retrofitting such systems for DS and DT functionality requires substantial investment in sensors, communication networks, hardware modifications, and interface bridging, often making implementation technically complex and cost-intensive [191].

Lastly, economic considerations also limit the establishment of DS or DT frameworks for membrane processes. These frameworks require investment in sensing hardware, data acquisition systems, computing infrastructure, cybersecurity protection, and skilled personnel capable of integrating membrane science with advanced analytics. While DS offers a lighter, one-directional digital layer, DTs require continuous bi-directional synchronization and automated model updating, significantly increasing computational and operational demands. For many small and medium-sized membrane plants, the cost–benefit balance does not yet justify deployment [192].

Overall, the implementation of DS and DT in membrane processes shares many of the challenges faced by AI and IoT. These challenges include data quality issues, infrastructure requirements, privacy and cybersecurity concerns, trust limitations, and unrealistic performance expectations. However, DTs face unique barriers, including the absence of standardized modeling workflows, high economic burden, and domain-specific compatibility constraints. Overcoming these limitations is essential to unlocking the full potential of DS- and DT-enabled membrane systems in future industrial practice [193].

6. Future perspectives and research opportunities

- **Next-generation smart membranes**

Recent advances in membrane science have enabled the development of smart or stimuli-responsive membranes with tunable separation performance. Unlike conventional membranes, which suffer from fixed pore structures, limited functional adjustability, and persistent surface or internal fouling, these next-generation materials exhibit dynamic, reversible changes in response to external triggers [194]. By incorporating responsive polymers, nanomaterials, or functional groups into membrane substrates, these systems can modulate their physicochemical properties in response to environmental or operational cues, including temperature, pH, ionic strength, light, magnetic fields, redox conditions, or specific biomolecules. Such multifunctional architectures

Table 8
Advantages and disadvantages of phenomenological, data-driven, hybrid modeling, DS, and DT.

Approach	Advantages	Limitations
Phenomenological (Physics-Based) Modeling	• Based on first principles and theoretical foundation • Provides interpretable outcomes • Respects conservation laws • Efficient in data use • Allows systematic parameterization	• Requires strong domain expertise • The development process is complex • Needs multidisciplinary effort • Solutions can be mathematically challenging • Time and resource-intensive • Limited scalability
Data-driven modeling	• Does not require prior domain knowledge • Straightforward development • Captures nonlinear system dynamics • Detects hidden patterns in data • Enables rapid prototyping	• Highly data-dependent • Risk of over-parameterization • Prone to overfitting • Can violate physical conservation laws • Often lacks interpretability
Hybrid modeling	• Combines physical principles with data insights • Improves interpretability • Enhances data efficiency • Capable of modelling complex system dynamics • Preserves conservation laws	• Requires some domain knowledge • Needs expertise in data-driven methods • Implementation complexity • Interpretability may decrease if poorly integrated
Digital Shadow	• Lightweight and easy to deploy • Low computational load • Practical for existing plants • Supports decision making.	• Not bi-directional • Cannot perform control or full optimization • Depends on sensor quality • Very few implementations in membrane systems (major gap).
Digital Twin	• Supports advanced automation and predictive control in principle; • Most comprehensive representation • Supports optimization and maintenance • Highest predictive power.	• Full autonomous closed-loop operation has not yet been demonstrated in industrial membrane plants • Requires extensive sensing and infrastructure • Complex integration • Computationally demanding • Only emerging in membrane research (significant gap).

offer tunable permeability and enhanced selectivity while improving antifouling behavior, mechanical robustness, pollutant removal efficiency, and compliance with effluent quality standards [195,196]. Their durability, reversibility, and controllability provide a promising pathway toward more adaptive and energy-efficient membrane operations.

Importantly, when integrated with DS or DT frameworks, stimuli-responsive membranes can enable self-adaptive separation. Incorporating embedded sensors and simple actuators (e.g., electrical triggers, automated valves, backwash/air-scour mechanisms) allows AI-driven models to update operating conditions in real time, maintaining flux and selectivity under fluctuating feed conditions. Recent work highlights rapid progress in innovative and semi-adaptive membrane materials and outlines how their intrinsic responsiveness can be coupled with data-centric, ML-enabled control layers, providing a materials platform capable of supporting next-generation DS- and DT-enabled autonomy [197,198].

- **Advancements in AI and ML for DT optimization**

Data-driven models are already improving fouling prediction, flux recovery scheduling, and operating-point optimization in membrane operations. The next step is closed-loop autonomy, resulting in a system that can monitor, decide, and act in real time without external/manual intervention. Attempts have been made regarding this goal. For instance, Moller et al. [43] developed a data-driven DT for water UF that demonstrated real-time control with explicit uncertainty handling. This study serves as a model for DT optimization in separations. Moreover, reinforcement learning (RL) studies in membrane capacitive deionization and full-scale MBR, such as the work of Shim et al. [199], have shown that self-learning controllers can reduce aeration/energy use while stabilizing effluent quality, foreshadowing autonomous membrane operation when embedded in DTs. Broad reviews of AI for membranes map the algorithmic toolbox (ML, RL) and emphasize hybrid surrogates to preserve physical consistency while enabling fast inference [200–203]. However, as noted, this is a novel case study that warrants further attention in the years ahead.

- **DS and DT–driven sustainability**

DSs and DTs offer powerful routes for advancing sustainability within membrane technologies. By enhancing forecasting accuracy, enabling soft sensing of latent or unmeasured states, such as fouling resistance, concentration polarization, or aging-related performance decay, and supporting optimization under uncertainty, DSs and DTs can substantially reduce energy consumption and material use while minimizing operational waste. In membrane processes, data-driven DSs and DTs enable more stable operation at constant production rates, resulting in observable sustainability benefits. Beyond operational improvements, DT-enabled analyses are increasingly connected to circular-economy strategies that transition membrane systems from the traditional linear model of resource extraction, production, and disposal toward extended use, reuse, and recovery.

. DTs enable continuous tracking of equipment conditions throughout the life cycle, thereby supporting predictive maintenance, end-of-life diagnostics, and informed decisions regarding the remanufacturing or refurbishment of membrane modules, housings, seals, and pumps. This provides a practical blueprint for transforming membrane assets into circular value chains, thereby improving resource efficiency and reducing waste streams [43,204]. Moreover, recent studies quantify these advantages at the system scale, showing that DT-guided operations can reduce waste generation, extend membrane lifetime, and cut energy consumption across water treatment and gas-separation applications. Collectively, the emerging evidence indicates that integrating DTs with advanced ML architectures will be pivotal to achieving carbon-neutral, resource-efficient membrane systems in the coming decades [205].

To fully harness hybrid modeling, DSs, and DTs in membrane science, collaborative research efforts are essential. The community should share datasets, adopt standardized validation methods, and develop open-source DS and DT tools. By enhancing sensors, integrating mechanistic and ML models, and creating scalable frameworks, researchers can transition from conceptual models to practical, reliable, and sustainable membrane systems.

7. Conclusion

This review has evaluated recent advances in hybrid modeling, DSs, and DTs of membrane technologies applied across several industries, desalination and water treatment, MBRs, gas separation for CO_2 and H_2,

O_2/N_2 separation, CMRs, and PEMFCs. The contributions of CFD in generating high-resolution synthetic datasets, strengthening mechanistic understanding, and supporting the development of data-driven and hybrid models have been elaborated. The examined case studies demonstrate significant progress in predicting fouling and key operating parameters, improving fault detection in PEMFC systems, and enabling real-time decision making through DT frameworks. Overall, hybrid models facilitate high-fidelity prediction, DSs provide continuous monitoring, and DTs offer a pathway toward intelligent, self-optimizing, and sustainable membrane-based separation systems. These all contribute to membrane health and to efficient prediction and improvement by enabling the realization of data values.

Despite these advances, current applications remain limited, especially for DSs. Future work should focus on integrating DS frameworks into membrane separation systems, particularly where full DT infrastructure is not yet feasible. Research on DS and DT applications in gas separation, specifically H_2 purification and CO_2 capture, also remains limited. Expanding these efforts will better support sustainability and decarbonization goals. Key research gaps and technology shortfalls include:

- Insufficient sensor and data infrastructure for continuous, high-quality monitoring of membrane processes.
- Limited studies on DS implementations, especially in real-time membrane monitoring and visualization.
- Lack of scalable DT architectures for large membrane plants.
- Limited studies applying DS and DT to membrane-based gas-separation processes.
- Absence of standardized validation and benchmarking protocols for DS and DT performance in membrane separation processes.

To fully harness the potential of hybrid modeling, DSs, and DTs in membrane science, coordinated research efforts are urgently needed. The community should focus on shared datasets, standardized validation, open-source toolkits, and cross-disciplinary collaboration among membrane engineers, data scientists, and control specialists. By advancing sensor infrastructure, integrating mechanistic and ML models, and developing scalable DS and DT frameworks, researchers can accelerate the transition from conceptual digital models to practical, reliable, and sustainable membrane-based systems.

Acronyms

ADT	Adaptive Boosted Trees
AI	Artificial Intelligence
ANN	Artificial Neural Network
CAMD	Computer-Aided Molecular Design
CCS	Carbon Capture and Storage
CFD	Computational Fluid Dynamics
CMR	Catalytic Membrane Reactor
CNN	Convolutional Neural Network
DL	Deep Learning
DS	Digital Shadow
DT	Digital Twin
ETR	Extra Tree Regressor
GB	Gradient Boosting
GBR	Gradient Boosting Regressor
IoT	Internet of Things
KNN	K-Nearest Neighbor
MBR	Membrane Bioreactor
MCH	Methylcyclohexane
MD	Membrane Distillation
MF	Microfiltration
ML	Machine Learning
MMM	Mixed Matrix Membrane
MOF	Metal Organic Framework
MR	Membrane Reactor
NF	Nanofiltration
NNPC	Neural Network Predictive Control
OPC UA	Open Platform Communications Unified Architecture
PBMR	Packed Bed Membrane Reactor
PEMFC	Proton Exchange Membrane Fuel Cell
RL	Reinforcement Learning
RO	Reverse Osmosis
SCADA	Supervisory Control and Data Acquisition
SGMC	Stochastic Grey Box Modelling and Control
SMC	Sliding-Mode Controller
SVM	Support Vector Machine
SVR	Support Vector Regressor
TFF	Tangential Flow Filtration
TSS	Total Suspended Solids
UF	Ultrafiltration
XGBoost	eXtreme Gradient Boosting

(*continued on next column*)

(*continued*)

CRediT authorship contribution statement

Kamran Ghasemzadeh: Writing – original draft, Supervision, Conceptualization. **Mostafa Jafari:** Visualization, Data curation. **Tara Torabi:** Visualization, Formal analysis, Data curation. **Amirpiran Amiri:** Visualization, Formal analysis. **Adolfo Iulianelli:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Adolfo Iulianelli, Master Degree in Chemical Engineering and Ph.D. in Chemical and Material Engineering at the University of Calabria (Italy). Nowadays, he is Senior Researcher at the National Research Council - Institute on Membrane Technology of Italy. He is/was scientific responsible of various scientific projects, coordinating the activities focusing on the application of membrane engineering in the field of green chemistry, renewable energy, decarbonized H_2 generation, fuel processing, CO_2 capture and valorization, gas separation.

Iulianelli is author and co-author of more than 270 scientific products (h-index = 48 (SCOPUS), 50 (GOOGLE SCHOLAR)) as peer reviewed articles, patents, books, chapters book, etc., and has been included since 2020 in the "World's Top 2% Scientists" list compiled by Stanford University & Elsevier

Iulianelli has been Editor of various Elsevier's books about hydrogen, membrane engineering and renewable sources, and currently he serves as Associate Editor of Fuel Processing Technology (Elsevier), co-Editor in Chief of the International Journal of Membrane Science and Technology (Cosmos), Associate Editor of Frontiers of Chemical Engineering (Frontiers), Executive Guest Editor for International Journal of Hydrogen Energy (Elsevier), Renewable Energy (Elsevier), and Chemical Engineering and Processing Process Intensification (Elsevier), and editorial board member of several others.

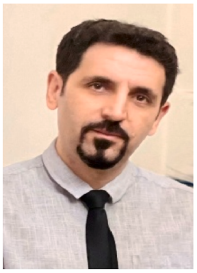
Iulianelli delivered several Plenary/Keynote and Invited lectures, serving as Chair/President and Member of numerous Scientific/Organizing Committees in international conferences, workshops and further scientific events.

He is President of the International Agency for Green Energy – Italian Chapter and served as Monitor Expert for the European Innovation Council and SMEs Executive Agency (EISMEA) and as Expert Independent Rapporteur for the European Climate, Infrastructure and Environment Executive Agency (CINEA), Expert Referee for MERANET agencies as well as for the US Department of Energy (DOE), the Fulbright Award Romania-U.S. Program, etc. and as Expert Evaluator for the Italian Ministry of Foreign Affairs and Commerce as well as for the Ministry of University and Research (ANVUR-VQR).

Iulianelli is qualified as Full Professor in "Chemical plants and technologies" & "Systems, methods and technologies of chemical and process engineering" and is Member of the "Industrial & Civil Engineering Ph.D. board" at University of Calabria (Italy), supervising several Post-doc researchers, and Master Degree students.

Main fields of research:

- Inorganic membrane reactors for decarbonised H_2 generation via reforming reactions of derived of fossil fuels and renewable sources.
- Pd/Pd-alloys, carbon, graphene oxide, polymeric membranes for H_2 separation/purification.
- CO_2 separation and conversion into added value products (bio & e-fuels) by membrane processes (polymeric, inorganic.)
- Energy Efficiency, Renewable Sources & Electrification



Kamran Ghasemzadeh holds a Ph.D. in Chemical Engineering from Sahand University of Technology, Iran, where he developed nanostructured silica membranes and integrated membrane reactors for hydrogen production. He also obtained his MSc and BSc degrees in Chemical Engineering from the same institution, specializing respectively in process design engineering.

Dr. Ghasemzadeh is currently a senior researcher at the University of Edinburgh, School of Engineering, Institute for Materials and Processes (UK), and previously served as senior researcher at the University of Manchester (2022–2024). Before moving to the UK, he was Associate Professor (2018–2022) and Assistant Professor (2015–2018) at the Chemical Engineering Faculty of Urmia University of Technology (Iran), where he also held several leadership positions, including Head of Department and Head of the Research and Innovation Institute.

Dr. Ghasemzadeh has authored 81 peer-reviewed journal articles, 39 Elsevier book chapters, and has contributed as editor to 12 books published by Elsevier and Wiley. His work has received 1973 citations, with an H-index of 27 and an i10-index of 55, reflecting strong impact in hydrogen production, membrane engineering, process intensification, and computational modelling. He has delivered more than 31 presentations at national and international conferences, including seven invited talks.

He currently serves as Associate Editor for Frontiers in Membrane Science and Technology and is an editorial board several scientific journals. He has been a Guest Editor for special issues on hydrogen production, membrane processes, and computational methods in journals such as the International Journal of Hydrogen Energy, Membranes and Process journals. Dr. Ghasemzadeh is also a member of the UKRI Peer Review College and has reviewed extensively for high-impact journals and major publishers including Elsevier and Wiley.

As an educator, Dr. Ghasemzadeh has a strong teaching portfolio spanning undergraduate and postgraduate modules such as Unit Operations, Process Control, Thermodynamics, Fluid Mechanics, Membrane Technology, and Advanced Modelling. He has supervised multiple PhD and MSc students across the UK and Iran, and developed computational training materials widely used in membrane process design education.

Main Fields of Research:

- Membrane and membrane-reactor engineering for hydrogen production, CO_2 capture, and intensified chemical processes
- Design and fabrication of inorganic, Pd-based, graphene oxide, and advanced polymeric membranes
- Digital Twins/Digital Shadow models, hybrid ML–CFD modelling, and physics-informed machine learning
- Process intensification, sustainable fuel production, and clean energy pathways
- Techno-Economic Analysis (TEA) and Life Cycle Assessment (LCA) for feasibility evaluation of net-zero processes
- Advanced computational approaches for optimization and simulation across chemical engineering systems



Dr. Amirpiran Amiri is a Senior Lecturer in Chemical Engineering at Aston University and an Honorary Associate Professor at the University of Birmingham. He is an internationally recognized researcher in hydrogen, fuel cells, and electrolysis, specializing in process systems engineering and digital innovation. He leads the Hydrogen and Fuel Cell Research Group at Energy and Bioproducts Research Institute (EBRI) at Aston where he co-founded the Wolfson Low Carbon Hydrogen Centre. Dr Amiri acts as Deputy Chair of Energy Research Accelerator Hydrogen Specific Group. He led 14 research projects, and he has been Aston's PI in large projects such as HyDEX (Research England funded) and METASIS 2.0 (EPSRC funded). He has published more than 42 papers, supervised 6 PhDs & 4 Postdocs, received several research awards such as the John Brodie Medal and Hydrogen Award 2025, and developed 2 pioneering MSc programmes in digital and sustainable chemical engineering. Dr Amiri's member of EPSRC Review College, ERA Research Committee.

Main Fields of Research:

- Hydrogen and Ammonia
- Fuel Cells and electrolysis
- Process modelling and simulation, digital process engineering
- Renewable energies
- Decarbonization



Mostafa Jafari received his Ph.D. in Chemical Engineering from the University of Tehran, where he focused on molecular and fluid-dynamics modeling of braid-reinforced hollow-fiber membranes for advanced wastewater treatment. He is currently a Senior Researcher at the Institute of Liquefied Natural Gas (LNG) and a Postdoctoral Fellow at the University of Tehran. His research centers on cryogenic carbon-capture technologies and their comparison with conventional and membrane-based processes, with broader expertise in Aspen HYSYS simulation, membrane engineering, cryogenic separation, CO_2 capture, and net-zero flare-gas valorization strategies.

He has authored and co-authored scientific publications across high-impact journals such as Fuel Processing Technology, Energy Conversion and Management, Green Chemistry, and Journal of Water Process Engineering. His portfolio includes contributions to membrane-based gas separation, CFD modeling of catalytic and reinforced hollow-fiber membranes, cryogenic CO_2 separation, and integrated GTL/PtX pathways.

According to Google Scholar, he has accumulated 149 citations, with an h-index of 7 and i10-index of 5, reflecting his growing scholarly influence in chemical engineering and sustainable energy systems.

Jafari's recent works include:

- Integrated environmental and techno-economic assessment of membrane-based GTL processes for flare-gas valorization.
- CFD and molecular-level modeling of braid-reinforced hollow-fiber membranes.
- Techno-economic assessment of bio-based routes for acrylic acid.
- Membrane-assisted ethylbenzene dehydrogenation for intensified styrene production.

Over recent years, he has contributed to the development of hybrid energy systems, polygeneration platforms, and flare-gas-to-fuel routes that support national goals in emission reduction, energy efficiency, and the valorization of wasted hydrocarbons. His academic work includes collaboration with research groups in membrane technology, sustainable gas processing, and hydrogen production, contributing to projects on Pd-based membrane reactors, MOF/polymer membrane systems, cryogenic separation units, and renewable-integrated GTL pathways.

Jafari continues to contribute to the advancement of sustainable gas processing and low-carbon technologies, with an emphasis on integrating membrane science, reaction engineering, and process intensification into scalable industrial pathways. His ongoing work aims to bridge the gap between simulation, materials innovation, and commercial-scale deployment of membrane-enabled processes.

Main fields of research:

- Membrane-based gas separation: polymeric, inorganic, and mixed-matrix membranes
- Techno-economic and environmental assessment of energy systems
- CFD and molecular simulation of hollow-fiber membranes
- Flare-gas valorization (GTL, FGTG, power generation)
- Hydrogen production and purification
- Polygeneration systems and renewable-integrated fuel synthesis



Tara Torabi holds a Bachelor's and Master's degree in Chemical Engineering from Urmia University of Technology (Iran), where she worked on membrane reactors, hydrogen production pathways, and advanced computational modelling. She will begin her PhD in Chemical Engineering at the University of Manchester (UK) in January, focusing on AI-assisted clean-energy systems and digitally enabled reactor technologies.

Torabi has contributed to 8 scientific publications, including peer-reviewed journal articles and book chapters, covering research areas such as hydrogen production, process intensification, and hybrid ML–CFD modelling.

Main Fields of Research:

- Membrane and membrane-reactor technologies for hydrogen and clean-fuel production.
- The application of AI-enhanced modelling frameworks in chemical engineering, including Digital Twin/Digital Shadow development, hybrid ML–CFD approaches, and advanced machine-learning techniques

ustainable and renewable energy pathways, hydrogen and biohydrogen production, and technologies supporting the transition beyond fossil fuels

- Computational optimization, advanced reactor modelling, and data-driven process engineering enabled by artificial intelligence